

How Do We Know They Work? A Systematic Review of Evaluation Practices and Methodological Rigor in LLM-Based Agentic Educational Systems

David Henderson Ph.D.¹

¹ Stanford University, Stanford, CA, USA

Correspondence: David Henderson, dhenders013@proton.me

ORCID: 0000-0003-3523-0175 (<https://orcid.org/0000-0003-3523-0175>)

(Single author.)

Abstract

Background. Large language model (LLM) based *agentic* educational systems, which plan, use tools, reflect, or coordinate as multiple agents, are proliferating, but whether they are evaluated rigorously enough to support claims about learning is unclear.

Objective. To systematically characterize the evaluation practices and methodological rigor of LLM-based agentic educational systems, and to derive a reporting standard for the field.

Methods. Following PRISMA 2020, I searched eight sources (Scopus, Web of Science, ACM, IEEE, ERIC, ACL Anthology, arXiv, EdArXiv) for empirical studies of agentic educational systems published from January 2023 to June 2026. A single reviewer screened 3,710 deduplicated records, assisted by a validated advisory LLM triage (blind human-AI agreement Cohen's $\kappa = 0.65-0.74$) that recommended but never determined decisions. Openly accessible full text was required for inclusion: of 477 records eligible at title/abstract, the 132 meeting that criterion were coded against a 41-field rigor-and-reporting codebook; 129 agentic studies formed the analytic set.

Results. Evaluation was predominantly output-oriented: 56.6% used no human learners and only 23.3% measured an actual learning outcome. Rigor and reproducibility were limited: 1.6% used fully validated instruments, 2.3% controlled for novelty effects, 17.8% were fully reproducible, and 50.4% released prompts. Appraisal-critical reporting was sparse (ethics/IRB unreported in 68.2%). Studies disclosed most items (mean 16.2/18) but met far fewer in practice.

Conclusions. The field is closer to a transparency norm than a rigor norm. I propose the Agentic Educational AI Evaluation and Reporting (AEER) checklist, a disclosure-focused reporting standard targeting these gaps at low cost. Findings characterize the openly accessible agentic-education literature.



Keywords: large language models; agentic AI; educational technology; intelligent tutoring systems; systematic review; PRISMA; evaluation; methodological rigor; reproducibility; reporting standards

1. Introduction

Generative artificial intelligence has moved through education faster than almost any prior instructional technology, and the frontier has already shifted again. The first wave of classroom research centered on single-turn use of conversational models such as ChatGPT for drafting, explanation, and question answering. The current wave is *agentic*: large language models (LLMs) embedded in systems that plan, use external tools, reflect on and revise their own outputs, or coordinate as multiple specialized agents toward an instructional goal (Kamalov et al., 2025). These systems now take on roles once reserved for human educators or for hand-engineered intelligent tutoring systems (ITS) by diagnosing skill gaps and sequencing personalized learning paths (e.g., the multi-agent GenMentor framework; Wang et al., 2025), generating adaptive formative feedback (Stamper et al., 2024), acting as “teachable” peers, and supporting teachers with curriculum and assessment generation. The pace of development is reflected in a rapidly accumulating primary literature and in synthesis work that catalogs what such systems can do, including a recent capability survey of LLM agents for education (Chu et al., 2025).

Yet capability is not evidence of effectiveness. The central question for any educational technology is not whether it can be built but whether it can demonstrably help people learn. It is here that the agentic-education literature is far less mature than its rate of publication suggests. Adoption and enthusiasm are outpacing the quality of the evidence base, and the gap is rarely examined head-on.

Three recurring weaknesses are visible across primary studies. First, **outcomes are frequently measured by proxy**. Many evaluations report the quality of the system’s *outputs*. These are rated by experts or, increasingly, by other LLMs acting as judges, or by learner *satisfaction and perceived usefulness*, rather than measured changes in knowledge, transfer, or retention. Reviews of LLM-based feedback have noted that very few studies employ controlled pre/post designs or learning analytics to establish actual learning effects (Stamper et al., 2024). Second, **study designs are often underpowered and decontextualized**. Evaluations are commonly conducted with small convenience samples in laboratory or one-shot conditions rather than authentic classrooms over meaningful durations. A representative example evaluated an LLM tutoring system with twelve high-school participants (Jiang & Jiang, 2024). Comparators, where present, are frequently a vanilla LLM rather than an established instructional baseline. Third, **theoretical grounding and reproducibility are inconsistent**. Current LLM classroom systems are often noted to lack the explicit learning-science foundations that anchored earlier ITS work (Cohn et al., 2026), and reporting of base models, versions, prompts, and code remains uneven, undermining replication. Compounding these issues, ethical and oversight considerations, including hallucination, bias, human-in-the-loop control, and safeguards when learners are minors, are unevenly addressed even as deployment accelerates.

Existing review literature does not close this gap. Capability-oriented surveys map the design space of LLM agents in education but do not appraise evidence quality (Chu et al., 2025). A systematic review of AI agents in computer-supported collaborative learning addresses one specific pedagogical context (Ba et al., 2026). And a meta-review has even consolidated dozens of existing generative-AI-in-education reviews (Zhang, Deng, & Shadiev, 2026), underscoring how saturated the *descriptive* synthesis space has become. What is missing is a systematic, cross-cutting appraisal of *how rigorously agentic educational systems are actually evaluated* that examines the designs used, the constructs measured, the validity and reproducibility of the evidence, and the systematic weaknesses that follow. Without such an appraisal, the field cannot distinguish demonstrations from durable findings, and practitioners and policymakers inherit claims whose evidentiary basis is unclear.

This review addresses that gap. Following PRISMA 2020, I systematically identify empirical studies of LLM-based agentic and multi-agent educational systems published since 2023 and characterize their evaluation practices along design, outcome, rigor, theoretical-grounding, reproducibility, and oversight dimensions. I ask: (RQ1) what evaluation designs are used and how have they evolved; (RQ2) what outcome constructs are measured and in what combinations; (RQ3) what is the methodological rigor and reproducibility of the evidence base; and (RQ4) what minimum standard would meaningfully raise evidence quality. My contribution is threefold: (1) the first systematic map of evaluation rigor in agentic educational AI, (2) an evidence-quality synthesis exposing where the field's weaknesses are concentrated, and (3) a proposed reporting and evaluation standard: the Agentic Educational AI Evaluation and Reporting (AEER) checklist, intended to move the field from demonstration toward credible, cumulative evidence.

2. Methods

2.1 Reporting standard and protocol

This review was conducted and is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement (Page et al., 2021). The review was not preregistered; the search strategy, eligibility criteria, and extraction codebook were specified before screening began and are reported here and in the supplementary repository to support scrutiny and reuse. Because the object of this review is the *quality of evaluation evidence* rather than a pooled treatment effect, I adopt a structured descriptive synthesis rather than a meta-analytic approach (Section 2.8).

The review is scoped to studies whose full text is openly accessible (Section 2.2). This criterion was adopted for verifiability: it ensures that every study in the analyzed corpus can be independently obtained and re-appraised by any reader, which carries particular weight for a review whose object is reporting practice, since the evidence base for the claims made here is itself fully open to inspection. What that scope does and does not license by way of generalization is stated explicitly throughout and addressed in Sections 2.6 and 2.9.

2.2 Eligibility criteria

Eligibility was defined a priori using an adapted PICOS framework.

Population. Learners and/or educators in any formal or informal educational setting, at any level (K-12, higher education, professional, or informal learning).

Intervention. LLM-based systems exhibiting at least one *agentic* property: autonomous planning, external tool use, reflection or self-correction, or multi-agent collaboration, applied to an explicit educational function, including but not limited to tutoring, formative feedback, learner mentoring, teachable-agent interaction, and teacher support (e.g., curriculum or assessment generation). Systems built on rule-based or pre-LLM architectures were excluded. Single-turn conversational use of an LLM with no agentic property was excluded from the analytic corpus but recorded as a *boundary category* to document the agentic/non-agentic threshold (Section 2.7).

Comparator. Any or none. Comparator type was extracted as a variable rather than used as an inclusion criterion.

Outcomes. No specific outcome was required for inclusion. Because the review characterizes *which* outcomes studies report, all reported outcome types were eligible and coded (Section 2.7).

Study/source types. Empirical primary studies reporting some form of evaluation. I included peer-reviewed journal articles and full papers in archival conference proceedings, and I screened and coded preprints (arXiv, EdArXiv) and openly archived conference material (ACL Anthology), flagging non-peer-reviewed records so that venue was recorded as a study characteristic. I excluded editorials, opinion and position pieces, purely architectural descriptions with no evaluation, and secondary studies (reviews, meta-analyses).

Timeframe. January 2023 to June 2026, reflecting the emergence of LLM-based agentic educational systems.

Language. English-language reports only, acknowledged as a limitation (Section 2.9).

Accessibility. Openly accessible full text was required for inclusion. Reports whose full text could not be obtained through open-access routes (Section 2.6) were not eligible; these records were assigned exclusion code E8 at the full-text stage rather than at screening. This criterion was adopted for verifiability: it ensures that every analyzed study can be independently retrieved and re-appraised by any reader, so that the evidence base of a review concerned with reporting practice is itself open to inspection. Its consequences for representativeness are addressed in Sections 2.6 and 2.9.

Exclusion codes. At title/abstract screening, excluded records were assigned a reason code from E1 to E7: E1 not agentic; E2 not educational; E3 no evaluation reported; E4 wrong publication type (incl. secondary studies); E5 not English; E6 outside timeframe; E7 duplicate. A further code, E8 full text unavailable, was applied only at the full-text stage (Section 2.6); records assigned E8 are therefore not among those excluded at screening.

The screening rule under uncertainty was inclusive (uncertain records advanced rather than being excluded).

2.3 Information sources

I searched eight sources: Scopus, the Web of Science Core Collection, the ACM Digital Library, IEEE Xplore, ERIC, the ACL Anthology, arXiv, and EdArXiv. Bibliographic-database searches (Scopus, Web of Science, ACM, IEEE, ERIC) were executed through Stanford University institutional access. arXiv, EdArXiv, and the ACL Anthology were searched programmatically (via the OpenAlex and native arXiv / OSF / ACL Anthology interfaces). The search was executed on 14 June 2026; the full per-source query strings, export dates, and record counts are provided in the supplementary repository.

I supplemented database and source searching with backward and forward citation chasing on the included studies and on capability surveys and reviews identified during scoping.

2.4 Search strategy

The search combined three concept blocks: (i) LLM / foundation model, (ii) agentic system, and (iii) education; joined with AND, and adapted to the syntax and field tags of each source. A representative form of the strategy is:

```
("large language model*" OR "LLM" OR "generative AI" OR "GPT" OR "foundation model*")
AND ("agent" OR "agentic" OR "multi-agent" OR "multi agent" OR "pedagogical agent"
      OR "tutoring system" OR "teachable agent" OR "AI tutor")
AND ("education" OR "learning" OR "teaching" OR "instruction" OR "student*"
      OR "classroom" OR "curriculum" OR "assessment")
```

The strategy deliberately contained **no evaluation-related terms**. Filtering on “evaluation” or “learning outcomes” would bias the corpus toward studies that already foreground rigorous evaluation, defeating the review’s purpose; evaluation practice was instead assessed analytically across all retrieved agentic-educational-system studies.

Two source-specific adaptations are noted for transparency. In ERIC, an education-only database, the education concept block was dropped as redundant and the LLM and agentic blocks were combined. For the programmatic sources (arXiv, EdArXiv, ACL Anthology), retrieval used local keyword filtering over harvested metadata and records rather than native Boolean field search; this difference, and its sensitivity implications, is documented in the repository run log. Database-specific field constraints (e.g., abstract-field versus full-text search, wildcard and term limits) are recorded per source.

2.5 Study selection and screening

Deduplication. Records from all sources were assembled and deduplicated using a layered rule: exact DOI match; then title similarity ≥ 0.92 within the same year; then title similarity ≥ 0.97 across years, restricted to pairs in which at least one record originated from a preprint server (arXiv / EdArXiv), so that preprint-publication pairs were reunited

without over-merging distinct papers. Where a preprint and a published record were identified as the same work, the published record was retained as canonical and the preprint identifier preserved on the surviving record. This produced **6,120 records identified, 2,410 duplicates removed, and 3,710 unique records** taken forward to screening.

Single-reviewer design with AI-advisory triage. Screening was performed by a single reviewer. To assist, but not replace, the reviewer, I used a large language model as an advisory “second screener” whose suggestions were recorded but never determinative. I (the human reviewer) made the inclusion decision on every record, and the model excluded nothing autonomously. Records were managed and decided in Rayyan (Ouzzani et al., 2016). I describe the AI-advisory procedure in full because it is a methodological feature of the review, not an incidental tool.

For each record, the model received only the title, abstract, and a fixed eligibility prompt encoding the criteria of Section 2.2, and returned a structured advisory output: a label (include / maybe / exclude), a confidence level, an exclusion reason code, and a short rationale. The model was given no bibliographic metadata, no venue, and no human decision, so that its suggestion could not be anchored by information outside the title and abstract.

Configuration selection. Before full deployment, the reviewer screened a seeded random sample of 50 records unaided to serve as a gold standard, and four model configurations: two models (Claude Sonnet 4.6 and Claude Opus 4.8), each with extended “thinking” enabled and disabled, were run against that sample. Configurations were compared on agreement with the human decisions and, as the governing criterion, on *costly false-exclusions* (records the human included that the model would exclude). Extended thinking did not improve agreement and increased costly false-exclusions for both models; the Sonnet, thinking-disabled configuration produced no costly false-exclusions on the calibration sample and was the most decisive and least costly, and was selected for the full run. The configuration, the frozen prompt (with a recorded SHA-256 hash), and run provenance are logged in the repository.

Reliability (human-AI agreement). To quantify the reliability of the advisory model against the human screen, the reviewer screened a seeded random validation subset of 371 records (~10% of the unique corpus) **blind to the model’s labels**, in an isolated review containing no AI output. On this blind subset, human-AI agreement was Cohen’s $\kappa = 0.653$ for the three-class decision (include / maybe / exclude) and $\kappa = 0.743$ for the two-class decision (advance [include or maybe] versus exclude). I report this explicitly as *human-AI* agreement and not as inter-rater agreement between two human reviewers. The model’s disagreements were concentrated in the “maybe” band and on borderline agentic/eligibility cases; on the validation subset it produced three strict false-exclusions (human-included, model-excluded), each a borderline case that the human-owned decision recovered.

Full corpus screening. The selected configuration was then applied to the remaining records to generate advisory labels, which were imported into the main review as a non-binding column. The reviewer screened all records the model advanced (advised

include or maybe) in full; screened all records the model excluded with low confidence in full; and, for the large block the model excluded with high confidence, audited a seeded random sample of 300 in full before confirming the remainder of that block; every record thus received a human decision, and the high-confidence-exclude block was confirmed by audit rather than accepted unread. Records that the screen resolved to “maybe” and that the reviewer did not subsequently confirm as meeting the eligibility criteria were excluded; and the conservatism of this rule at the eligibility boundary is noted as a limitation (Section 2.9).

Outcome of screening. Of the 3,710 unique records screened, **477 were confirmed as included** at title/abstract. Reasons for exclusion were logged by code and are summarized in the PRISMA 2020 flow diagram (Figure 1).

Full-text eligibility. Full-text eligibility confirmation and data extraction were conducted as a single pass rather than as two separate stages. Inclusion was confirmed at title/abstract; the no-evaluation criterion (E3) and the agentic-property criterion (E1) were then verified against the full text during extraction, with any study failing either on reading dropped and recorded (three were dropped as non-agentic boundary cases; none under E3, see Section 2.7). This merged design departs from a conventional two-stage screen and is noted as a limitation (Section 2.9).

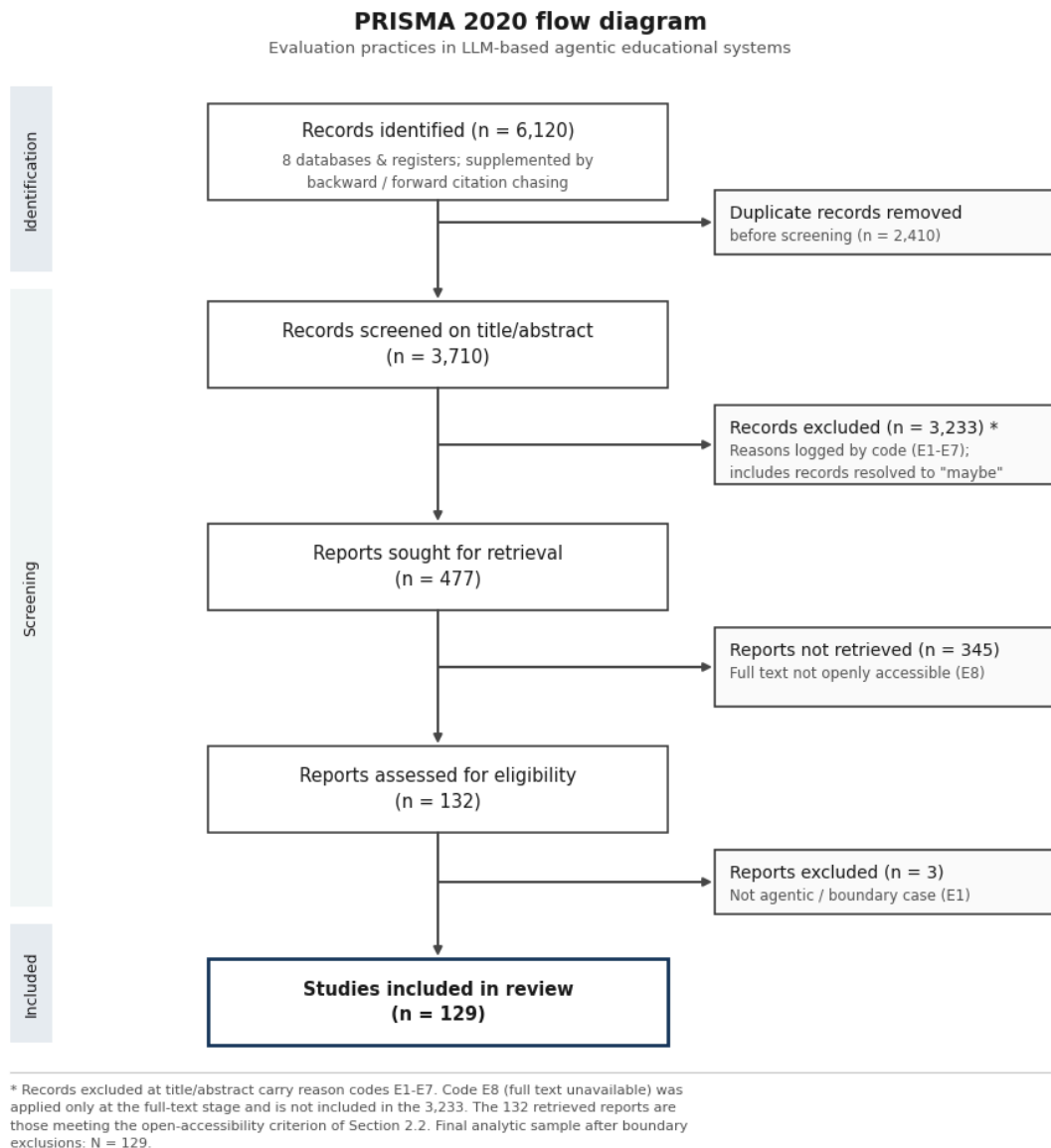


Figure 1. PRISMA 2020 flow diagram for the review. Records were identified from eight databases and registers ($n = 6,120$) and supplemented by citation chasing; 2,410 duplicates were removed before screening, leaving 3,710 unique records. A single reviewer screened all records with non-binding AI-advisory triage, excluding 3,233 (reason codes E1-E7; the excluded total includes records resolved to “maybe” that were not confirmed as eligible). Full text was sought for the 477 records eligible at title/abstract; 345 did not meet the open-accessibility criterion and were not retrieved, leaving 132 reports assessed for eligibility at full text. Three were excluded as non-agentic boundary cases, yielding 129 studies in the analytic set.

2.6 Full-text retrieval and the analyzed corpus

Full text was sought for all 477 records eligible at title/abstract. Retrieval used open-access locating services (Unpaywall, OpenAlex), direct preprint-server downloads (arXiv, EdArXiv,

ACL Anthology), and supplementary open-access aggregators. Of these 477 records, **132 (28%) had full text retrievable through these open-access routes** and therefore met the open-accessibility criterion of Section 2.2; the remaining **345 did not meet it** (predominantly paywalled journal and conference records) and were therefore not eligible; these records were assigned exclusion code E8 at this stage. Because the corpus is defined by open accessibility rather than drawn at random from the topically eligible records, and because open-access status may correlate with venue and with reporting practices, the results characterize the openly accessible agentic-education literature rather than the whole of it; this scope is restated wherever results are presented and in Section 2.9.

2.7 Data extraction and coding

Extraction was performed by the single reviewer using a standardized 41-field codebook (provided in full in the repository), piloted on five studies and then frozen. The codebook spanned: identification; system and agentic characteristics (the four agentic properties, educational role, subject domain, education level, base model and version); study design; sample and setting; outcome constructs measured (output quality, user perceptions, behavioral/process, learning outcomes); comparator; measurement validity (validated instruments, pre/post measurement, novelty-effect control, blinding); theoretical grounding; reproducibility (model/version disclosure, prompt release, code/data availability, protocol replicability); ethics and oversight (human-in-the-loop, bias/hallucination mitigation, ethics/IRB approval, involvement of minors and safeguards); and appraisal fields. Each study was keyed by its corpus record identifier.

AI-assisted extraction under an evidence constraint. Extraction was AI-assisted with a design that confined the model to locating stated facts and reserved all judgment for the reviewer. Fields recording *stated facts* (e.g., base model and version, sample size, venue, disclosure of prompts or code) were drafted by the model (Claude Sonnet 4.6) under a mandatory evidence constraint: each value had to be accompanied by a verbatim supporting quotation and its location in the source, and where no supporting text existed the field was forced to “not reported.” The reviewer verified each drafted field against its quotation. Fields requiring judgment: design classification, comparator and setting classification, measurement-validity and reproducibility appraisal, the Mixed Methods Appraisal Tool ratings, and the four agentic-property determinations; were coded by the reviewer; for these fields the model supplied only located evidence passages and no suggested value, so that its output could not anchor the reviewer’s judgment. The extraction prompt was frozen and its hash recorded, and per-study extraction provenance is logged in the repository.

No formal extraction-reliability (double-coding) assessment was conducted. Because the review had a single reviewer, no inter-rater extraction statistic is available; this is disclosed as a limitation (Section 2.9).

Boundary exclusions and analytic N. Of the 132 extracted studies, three were coded as exhibiting no agentic property and were excluded from analysis as boundary cases, and none were dropped under E3. This yields an **analytic sample of N = 129 studies** for all results reported below.

2.8 Risk-of-bias appraisal and synthesis

Study-level appraisal. Methodological quality was appraised using the Mixed Methods Appraisal Tool (MMAT; Hong et al., 2018), selected because the corpus spans qualitative, quantitative, and mixed designs. Each study was assigned an MMAT stream and a count of criteria met (0–5).

Synthesis. Given the heterogeneity of interventions, outcome constructs, and measures, a meta-analysis of effects was neither feasible nor appropriate; the heterogeneity is itself treated as a finding. I conducted a structured descriptive synthesis: frequencies and cross-tabulations of design, outcome construct, and rigor and reproducibility indicators, organized around the four research questions, and consolidated the findings into the proposed reporting standard (the AEER checklist). I additionally scored each included study against the AEER items by recoding the extraction fields under fixed per-item rules (reported in the supplementary materials), distinguishing *disclosure* (whether the item is reported) from *practice-met* (whether the positive state holds); not-applicable items were excluded from each item’s denominator. Because the items were derived from the gaps measured in this corpus, these compliance figures illustrate the standard rather than independently validate it.

2.9 Limitations

Several limitations follow from the review’s design and scope. (1) **Single reviewer.** Screening and extraction were performed by one reviewer; there is no inter-rater agreement statistic. The advisory-model design and the blind human-AI agreement statistic (Section 2.5) partially characterize screening reliability but do not substitute for independent human double-screening, and no extraction-reliability assessment was conducted. (2) **AI assistance.** A language model was used to advise screening and to draft stated-fact extraction fields; this was mitigated by a human decision on every record, an evidence-quotation constraint with human verification on drafted fields, and human-only coding of all judgment fields, but residual risk remains and the approach is disclosed in full for scrutiny. (3) **Open-accessibility scope.** The corpus comprises the 132 of 477 topically eligible records (28%) whose full text was openly accessible; because open-access status may correlate with venue and reporting norms, the findings characterize the openly accessible literature rather than the whole of it, a caveat of particular weight for a review of reporting practice. (4) **Merged eligibility and extraction.** Full-text eligibility confirmation was carried out during extraction rather than as a separate stage. (5) **Excluded “maybe” records.** Records the screen resolved to “maybe” and not subsequently confirmed as eligible were excluded, which narrows the corpus at its boundary. (6) **English-only and source coverage** may omit relevant work, particularly non-English and grey-literature reports; citation chasing partially mitigates this. (7) **Snapshot.** The corpus is a defined snapshot to the search date. (8) **Interpretation.** Judgment-based fields (e.g., theoretical grounding, reproducibility appraisal) involve reviewer interpretation against the codebook rules.

3. Results

3.1 The analytic corpus

Of the 132 studies whose full text was retrieved and coded, three were found to exhibit no agentic property and were excluded as boundary cases, leaving **N = 129** studies in the analytic set. As a reminder of scope, these 129 are the openly accessible studies among the 477 records eligible at title/abstract; of these, 68 appeared in peer-reviewed venues and 61 are preprints, and the distributions below should be read as characterizing the openly accessible literature rather than the whole of it.

The systems were strongly agentic by construction of the inclusion criteria, but the *profile* of agentic capabilities was uneven. Planning was the most common property (119/129, 92.2%), followed by multi-agent coordination (104, 80.6%), tool use (98, 76.0%), and reflection or self-correction (95, 73.6%) (Table 1).

Table 1. Prevalence of individual agentic properties across the analytic corpus (N = 129).

Agentic property	n (Y)	% of N
Planning	119	92.2%
Multi-agent	104	80.6%
Tool use	98	76.0%
Reflection	95	73.6%

Systems tended to combine several capabilities rather than exhibit one in isolation: 45.0% (58/129) were coded as having all four agentic properties and 37.2% (48) had three, while only 4.7% (6) exhibited a single property. The prevalence of multi-property systems, set against the lower and unevenly reported rates for individual mechanisms (Section 3.5), indicates that systems are frequently described as broadly “agentic” without each claimed capability being separately substantiated (Table 2).

Table 2. Number of distinct agentic properties exhibited per system (N = 129).

# agentic properties	n systems	% of N
1	6	4.7%
2	17	13.2%
3	48	37.2%
4	58	45.0%

3.2 RQ1: Evaluation designs

Evaluation designs were dominated by approaches that do not measure effects on human learners. The single most common design was system-only or simulation evaluation (73/129, 56.6%); that is, evaluation conducted without human-learner participants. Designs capable of supporting causal inference about learning were rare: only two studies (1.6%) used a randomized controlled design and 11 (8.5%) a single-group pre/post design,

with quasi-experimental (19, 14.7%) and observational (24, 18.6%) designs making up the remainder (Table 3).

Table 3. Distribution of evaluation design types (N = 129).

Design type	n	% of N
System-only / simulation	73	56.6%
Observational	24	18.6%
Quasi-experimental	19	14.7%
Single-group pre/post	11	8.5%
RCT	2	1.6%

Most studies did include some comparison condition (99/129, 76.7% had a comparator other than “none”), but the comparators were predominantly internal rather than instructional: ablations (61, 47.3%) and vanilla-LLM baselines (25, 19.4%) together accounted for the large majority, while comparison against a human tutor (7, 5.4%) or traditional instruction (5, 3.9%) was uncommon. Thirty studies (23.3%) reported no comparator at all. The pattern indicates that, where comparison occurs, it more often establishes that an agentic configuration outperforms a simpler model than that the system improves on an educational alternative (Table 4).

Table 4. Comparator conditions used across studies (N = 129).

Comparator type	n	% of N
Ablation	61	47.3%
None	30	23.3%
Vanilla LLM	25	19.4%
Human tutor	7	5.4%
Traditional instruction	5	3.9%
Other	1	0.8%

3.3 RQ2: Outcome constructs

The constructs studies measured were concentrated on the system’s outputs rather than on learning. Output quality was measured in 92.2% (119/129) of studies, behavioral or process measures in 56.6% (73), and user perceptions in 41.9% (54). In contrast, **actual learning outcomes: knowledge gain, transfer, or retention, were measured in only 23.3% (30/129) (Table 5).**

Table 5. Outcome-construct families measured across studies (N = 129).

Outcome construct	n (measured)	% of N
Output quality	119	92.2%

Outcome construct	n (measured)	% of N
Behavioral / process	73	56.6%
User perceptions	54	41.9%
Learning outcomes	30	23.3%

Framed as the gap between evidence and claim, 76.7% of studies (99/129) measured only proxy constructs (output quality, perceptions, or process) without a direct learning measure, while no study measured none of the four construct families. Most studies measured one (33.3%) or two (28.7%) construct types; measuring three (28.7%) or all four (9.3%) was less common. The predominance of proxy measurement is the central construct-validity concern of the corpus: claims about educational benefit rest, in three-quarters of studies, on indirect indicators (Table 6).

Table 6. Number of distinct outcome-construct families measured per study (N = 129).

# construct types measured	n studies	% of N
1	43	33.3%
2	37	28.7%
3	37	28.7%
4	12	9.3%

3.4 RQ3: Methodological rigor and reproducibility

Measurement validity. Use of validated outcome instruments was rare and frequently unreported. Only 1.6% (2/129) of studies used fully validated instruments and 31.8% (41) partially validated ones (33.3% combined, the figure used for AEER item C3 in Section 3.6); 18.6% (24) used ad hoc measures, and validity was not reported at all in 48.1% (62), which was the single largest measurement-reporting gap (Section 3.5). Controls for novelty or expectancy effects were almost entirely absent: only 2.3% (3) reported any such control, 70.5% (91) reported none, and 27.1% (35) were silent. Pre/post measurement was reported in 15.5% (20) of studies and was coded not-applicable in 77.5% (100), consistent with the proxy-heavy outcome profile; blinding was reported where applicable in 14.0% (18) (Table 7).

Table 7. Methodological rigor indicators: validated instruments, novelty-effect control, pre/post measurement, and blinding (N = 129).

Indicator	Y	Partial	N	NR / NA
Validated instrument	2 (1.6%)	41 (31.8%)	24 (18.6%)	62 NR (48.1%)
Novelty-effect control	3 (2.3%)	n/a	91 (70.5%)	35 NR (27.1%)
Pre/post measurement	20 (15.5%)	n/a	9 (7.0%)	100 NA (77.5%)

Indicator	Y	Partial	N	NR / NA
Blinding	18 (14.0%)	n/a	16 (12.4%)	95 NA (73.6%)

Reproducibility. Disclosure was strongest for the base model identity: 87.6% (113) reported the model and version clearly enough to identify, leaving 12.4% (16) that did not. These 16 comprise the six studies (4.7%) silent on the base model altogether, which are the not-reported cases counted in Table 10, plus ten that named a model without a usable version string (for example an unversioned “GPT-4” or “ChatGPT”). Beyond model identity, transparency dropped: prompts were released by only 50.4% (65), and code or data by only 39.5% (51). The protocol-replicability appraisal is the corpus’s defining reproducibility finding: no study was judged fully non-reproducible, but **only 17.8% (23) were judged fully replicable, while 82.2% (106) had major reported gaps (“Partial”).** For systems whose behavior is defined largely by prompts and orchestration, the release of prompts in only half the corpus means the intervention itself is under-specified in a large share of studies (Table 8).

Table 8. Reproducibility and transparency disclosure across studies (N = 129).

Reproducibility indicator	Reported / Y	% of N
Model + version disclosed	113	87.6%
Prompts released	65	50.4%
Code / data available	51	39.5%
Protocol fully replicable	23	17.8%
Protocol partially replicable	106	82.2%

Oversight and ethics. Human-in-the-loop oversight was reported in 58.9% (76) of studies and bias or hallucination mitigation in 70.5% (91). Reporting of ethics or IRB approval was the largest single gap in the corpus: 68.2% (88) did not report any ethics or approval status, 2.3% (3) reported none obtained, and only 29.5% (38) reported approval. This figure is depressed by the prevalence of system-only/simulation studies, for which IRB approval is often not applicable; restricted to the 64 studies involving human participants, ethics status was reported by 53.1% and approval by 50.0% (see Section 3.6).

Appraisal (MMAT). Overall methodological quality was modest, with a mean MMAT rating of 2.36 of 5 across the corpus. Ratings clustered at the low-to-middle of the scale (4 studies at 1/5, 77 at 2/5, and 46 at 3/5; only 2 reached 4/5 and none 5/5). The corpus was dominated by the quantitative-descriptive stream (94/129), which carried the lowest mean rating (2.17), consistent with the prevalence of system-only and proxy-based evaluation (Table 9). MMAT stream assignment reflects the appraisal category appropriate to each study’s reported methods and does not map one-to-one onto the design coding of Table 3, in which two studies were coded as using a randomized design.

Table 9. Methodological quality by MMAT study-design stream: study counts and mean quality ratings.

MMAT stream	n	mean rating (/5)
Quantitative descriptive	94	2.17
Qualitative	8	2.38
Mixed methods	13	2.92
Quantitative non-randomized	13	3.00
Quantitative RCT	1	4.00
All	129	2.36

3.5 RQ4: Reporting completeness

Treating “not reported” as the union of explicit not-reported codes, unclear (“?”), and blanks (and keeping not-applicable separate), the corpus showed a steep gradient of reporting completeness across fields. Identification and system-characterization fields were near-completely reported, whereas the fields most central to appraising evidence quality were the most often missing. The largest gaps were ethics/IRB approval (not reported in 68.2%), instrument validity (48.1%), novelty-effect control (27.1%), and the naming of a guiding theory (25.6%, indicating that a quarter of studies were not anchored in a named learning or instructional theory). Base model and version was unreported in only 4.7% (Table 10).

Table 10. Reporting completeness; not-reported rates by field (N = 129).

Field	not reported	% of N
Ethics / IRB approval	88	68.2%
Validated instrument	62	48.1%
Novelty-effect control	35	27.1%
Theory named	33	25.6%
Minors involved (unclear)	14	10.9%
Base model + version	6	4.7%
Human-in-the-loop	5	3.9%
Bias / hallucination mitigation	3	2.3%

The reporting-completeness gradient is the empirical basis for the proposed reporting standard. The fields a reader most needs in order to judge whether an agentic educational system has been credibly evaluated including how outcomes were validated, whether novelty was controlled, whether ethics approval was obtained, what theory motivates the

design, are precisely those most often absent. The AEER checklist (Section 4) is derived directly from this gradient, with each item tied to a measured gap above.

3.6 Retrospective AEER compliance

To show that the proposed standard is computable and discriminating, I scored each study against the AEER items by re-coding the extraction fields (mapping rules in the supplementary materials). Two distinct quantities are reported and should not be conflated. *Disclosure* (whether a study is non-silent on an item) is meaningful only for items the study itself must report; for several items: whether a learning outcome was measured, a comparator used, prompts or code released, or the protocol fully reproducible, the value was determinable from the full text for every study, so “disclosure” is 100% by construction and the substantive figure is the *practice-met* rate (the positive state). For the remaining items, disclosure reflects the study’s own reporting and, by construction, equals one minus the not-reported rates of Section 3.5. Because the checklist items were themselves derived from the gaps measured here, these figures illustrate the standard’s application rather than independently validate it.

Disclosure compliance (study-reported items). On items whose reporting depends on the study, disclosure was high for system, sample, oversight, and limitations reporting but fell sharply for the appraisal-critical items, mirroring Section 3.5. Where an item also has a positive practice state, the practice-met rate is far lower than the disclosure rate. This is the central pattern of this section: studies more often *report* an item than *satisfy* it (Table 11).

Table 11. Retrospective AEER compliance for disclosure-dependent Essential items: disclosure and practice-met rates (N = 129).

AEER item (Essential)	Disclosed	Practice met
A2 educational role & level	99.2%	n/a
A3/E4 base model + version	87.6%	87.6%
B2 sample (N learners)	96.1%	n/a
C3 instrument validity	51.9%	33.3%
D2 novelty / Hawthorne control	72.9%	2.3%
F1 ethics / IRB status	31.8% (53.1% over human-subjects subset, n = 64)	29.5% (50.0% subset)
F2 human-in-the-loop	96.1%	58.9%
F3 bias / hallucination mitigation	97.7%	70.5%
F4 minors status (+ safeguards)	89.1%	n/a
G1 limitations	98.4%	n/a

Practice compliance (determinable items). For items whose status was determinable for every study from the full text, disclosure is 100% by construction and the finding is the practice rate, which is low on exactly the items that bear on reproducibility and learning evidence (Table 12):

Table 12. Retrospective AEER practice compliance for determinable Essential items (N = 129).

AEER item (Essential)	Practice met
D1 real comparator present	76.7%
E1 prompts released	50.4%
E2 code / data available	39.5%
C2 learning outcome measured	23.3%
E3 protocol fully replicable	17.8% (+ 82.2% partial)

(The structural items A1 agentic profile, B1 design type, and C1 constructs measured were determinable for essentially all studies and are not informative as compliance figures.)

Recommended (design-dependent) items, scored only over their applicable subset: blinding was met in 52.9% (18/34 applicable) and pre/post measurement in 69.0% (20/29 applicable).

Per-study disclosure. Scoring each study on how many of the 18 Essential items it was non-silent about, the mean was 16.2 of 18 (90.1%), but only 19.4% (25/129) disclosed all 18. Disclosure clustered tightly at the top of the range (16–18 items for 71.3% of studies), indicating that near-complete *disclosure* is achievable and common, and therefore, that the residual gaps are a reporting choice rather than an inherent difficulty (Table 13).

Table 13. Distribution of studies by number of Essential AEER items disclosed, of 18 (N = 129).

# Essential items disclosed (of 18)	n studies	% of N
12	1	0.8%
13	4	3.1%
14	7	5.4%
15	25	19.4%
16	35	27.1%
17	32	24.8%
18	25	19.4%

The contrast between high disclosure (mean 16.2/18) and low practice on the reproducibility and validity items (full replicability 17.8%, validated instruments 33.3% fully or partially and only 1.6% fully, novelty control 2.3%) is the core compliance finding:

the field is closer to a transparency norm than to a rigor norm, and the gap AEER targets is therefore reportable at low cost.

3.7 Summary of findings

Across the analyzed corpus, three patterns recur. First, evaluation is predominantly output-oriented and proxy-based: most systems are assessed without human learners (56.6% simulation/system-only) and only 23.3% measure actual learning. Second, internal validity and reproducibility are weakly established and weakly reported: validated instruments (1.6% fully), novelty control (2.3%), full replicability (17.8%), and prompt release (50.4%) are all uncommon, and the modal MMAT rating is 2/5. Third, the reporting gaps are concentrated in the appraisal-critical fields (ethics 68.2%, instrument validity 48.1%, novelty control 27.1% not reported), which both limits what can be concluded from the current literature and motivates the reporting standard developed next. These findings characterize the openly accessible literature and should be read with the scope caveat of Methods 2.6 and 2.9 in mind.

4. The AEER Checklist

How to use the checklist

For each item, an author or reviewer records one of: **Reported (✓)**, **Not reported (NR)**, or **Not applicable (NA, with brief reason)**. The intent is *transparency*, not a quality score: an item answered honestly as “N/NR” is compliant with AEER; silence is not. A study does not need to perform every practice, but it should **state** whether it did. The complete item set is specified section by section in Tables 14–20 (Sections A–G), and a blank grid for scoring an individual study is provided in Table 21.

Tiering. [E] Essential: the minimum every study should report. **[R] Recommended:** strengthens rigor/reproducibility; report where applicable.

Section A: System & agentic characterization

Table 14. AEER checklist, Section A: System and agentic characterization.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
A1	Agentic profile [E]	State which of the four agentic capabilities the system has: planning, tool use, reflection, multi-agent, and how each is realized.	Coverage is uneven: planning 92.2%, multi-agent 80.6%, tool use 76.0%, reflection 73.6%; 45.0% of systems exhibit all four, but the <i>mechanism</i> is often unstated.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
A2	Educational role & level [E]	Name the instructional function (tutor, feedback, teachable agent, teacher support, curriculum/assessment) and the target education level.	Role/level was determinable for all but one study (99.2% disclosed); standardizes cross-study comparison.
A3	Base model + version + access date [E]	Report the exact base model(s), version string, and date/endpoint of access.	The base model was disclosed clearly enough to identify in 87.6% (113/129); 12.4% (16/129) did not give a usable model identity, and versionless “GPT-4”/“ChatGPT” entries are common.
A4	Architecture & orchestration [R]	Describe agent roles, coordination/control flow, memory, and retrieval components sufficient to redraw the system diagram.	Multi-agent designs in 80.6% of systems, but coordination detail is frequently absent (see A1).

Section B: Evaluation design

Table 15. AEER checklist, Section B: Evaluation design.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
B1	Design type [E]	State the strongest design used (system-only/simulation, observational, single-group pre/post, quasi-experimental, RCT) and why it was chosen.	56.6% are system-only/simulation; only 1.6% (2/129) are RCTs and 8.5% single-group pre/post; design choice is rarely justified.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
B2	Human learners vs simulation [E]	State whether outcomes come from real human learners or simulated/LLM “student” agents; report N and sampling.	56.6% used system-only/simulation designs with no human learners (§3.2); evaluation on simulated students is easily conflated with evaluation on learners. Sample size (N learners) was reported in 96.1%.
B3	Setting & duration [R]	Report authentic-classroom vs laboratory vs simulated, and one-shot vs multi-session vs longitudinal exposure.	Setting and duration were coded for all studies (0% not reported).

Section C: Outcome constructs & measurement

Table 16. AEER checklist, Section C: Outcome constructs and measurement.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
C1	Constructs measured [E]	State explicitly which construct families are measured: output quality, user perceptions, behavioral/process, learning outcomes, and which are not .	Output quality 92.2% but learning outcomes only 23.3% (30/129); 76.7% rely on proxies only.
C2	Learning vs proxy claims [E]	If learning is claimed, report a learning measure (knowledge gain/transfer/retention); if only proxies are measured, frame	Only 23.3% measure an actual learning outcome; the gap between proxy evidence and learning claims is the dominant

#	Item [tier]	What to report	Evidence motivating it (N = 129)
C3	Instrument validity [E]	conclusions as proxy-level. State whether each outcome instrument is previously validated, partially validated, or ad hoc.	construct-validity risk. Validity was not reported in 48.1% (62/129); only 1.6% used fully validated and 31.8% partially validated instruments; the largest <i>measurement</i> reporting gap.
C4	LLM-as-judge disclosure [R]	If an LLM scores outputs, report the judge model+version, prompt, and any human-agreement check.	Output-quality evaluation (92.2%) frequently relies on LLM judges; judge provenance is inconsistently reported (ties to A3).

Section D: Comparators & internal validity

Table 17. AEER checklist, Section D: Comparators and internal validity.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
D1	Comparator [E]	State the strongest comparison condition (none / vanilla LLM / ablation / human tutor / traditional instruction / other) and justify it.	23.3% have no comparator; of those with one, most are ablations (47.3%) or vanilla-LLM (19.4%); only 9.3% compare against a human tutor or traditional instruction.
D2	Novelty / Hawthorne control [E]	Report whether novelty/expectancy effects were controlled (active control, sufficient exposure) or	70.5% report no control and 27.1% are silent (NR); only 2.3% report any novelty control.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
D3	Blinding [R]	Report whether participants/assessors were blinded where feasible, or why NA.	Where applicable (n = 34), blinding was reported in 52.9% (18); across all studies 73.6% were NA; applicability itself is often unstated.
D4	Pre/post measurement [R]	For learning claims, report pre- and post-intervention measurement.	Pre/post reported in 15.5% overall; among applicable studies (n = 29) 69.0% reported it; 77.5% NA; consistent with the proxy-heavy outcome profile (C2).

Section E: Reproducibility & transparency

Table 18. AEER checklist, Section E: Reproducibility and transparency.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
E1	Prompts released [E]	Release the full system/agent prompts (paper, appendix, or repo).	Only 50.4% release prompts. For prompt-defined agentic systems, the intervention is under-specified in half the corpus.
E2	Code / data available [E]	State whether code and/or data are publicly available and link them.	Only 39.5% make code/data available; 60.5% do not.
E3	Replication-sufficient protocol [E]	Provide enough procedural detail (settings, decoding params, pipeline, eval scripts) for an	No study was rated fully non-reproducible, but only 17.8% are fully replicable,

#	Item [tier]	What to report	Evidence motivating it (N = 129)
		independent team to reproduce the evaluation.	82.2% have major gaps (“Partial”). The core reproducibility finding.
E4	Model + version disclosed [E]	(See A3.) Disclose model identity precisely enough to reproduce.	12.4% (16/129) do not disclose a usable model identity.

Section F: Ethics & oversight

Table 19. AEER checklist, Section F: Ethics and oversight.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
F1	Ethics / IRB approval [E]	Report ethics/IRB approval status (approved / exempt / not obtained / NA) for human-subjects work.	Ethics/IRB status was unreported in 68.2% overall; the largest single reporting gap; restricted to the 64 human-subjects studies, 53.1% reported status and 50.0% reported approval.
F2	Human-in-the-loop [E]	State whether human oversight of system outputs/decisions exists in the deployment described.	58.9% report human-in-the-loop, 37.2% report none, 3.9% silent; should be explicit given autonomy claims.
F3	Bias / hallucination mitigation [E]	Report measures addressing hallucination, factual error, or bias (and their evaluation).	70.5% report some mitigation; 27.1% report none. For educational content, absence should be stated, not implied.
F4	Minors & safeguards [E]	State whether participants are minors and, if so, describe specific	Minors-involved was unclear (“?”) for 10.9% of studies; where minors are

#	Item [tier]	What to report	Evidence motivating it (N = 129)
		safeguards (consent, data handling, content filtering).	involved, safeguards are frequently unstated.

Section G: Appraisal & administrative

Table 20. AEER checklist, Section G: Appraisal and administrative.

#	Item [tier]	What to report	Evidence motivating it (N = 129)
G1	Limitations & risk of bias [E]	Report author-acknowledged limitations and threats to validity.	Reported by 98.4% of studies; standardizing aids appraisal (corpus MMAT mean = 2.36/5).
G2	Appraisal stream [R]	Identify the study's methodological stream (qualitative / RCT / non-randomized / descriptive / mixed) to support appraisal.	72.9% fall in the quantitative-descriptive stream (mean rating 2.17/5), indicating shallow appraisal headroom across the field.



Proposed minimum (Essential) set: one-page version

A study meets the AEER **Essential** bar if it reports, for each item, one of Reported / NR / NA:

- **A1** agentic profile · **A2** role & level · **A3** base model + version + access date
- **B1** design type (+ justification) · **B2** human vs simulated learners (+ N)
- **C1** constructs measured (and not) · **C2** learning vs proxy framing · **C3** instrument validity
- **D1** comparator (+ justification) · **D2** novelty/Hawthorne control
- **E1** prompts released · **E2** code/data availability · **E3** replication-sufficient protocol · **E4** model disclosure
- **F1** ethics/IRB · **F2** human-in-the-loop · **F3** bias/hallucination mitigation · **F4** minors & safeguards
- **G1** limitations & risk of bias

(18 Essential items. Retrospective corpus compliance against this set is reported in Results §3.6: mean 16.2/18 disclosed, 19.4% disclosing all 18.)

Blank compliance grid (per study)

Table 21. Blank AEER compliance grid for scoring an individual study.

AEER item	Reported (✓)	NR	NA (reason)
A1 agentic profile			
A2 role & level			
A3 model+version+date			
B1 design type			
B2 human vs simulated			
C1 constructs measured			
C2 learning vs proxy			
C3 instrument validity			
D1 comparator			
D2 novelty control			
E1 prompts released			
E2 code/data			
E3 replication protocol			
E4 model disclosed			
F1 ethics/IRB			
F2 human-in-the-loop			
F3 bias/hallucination			
F4 minors & safeguards			
G1 limitations			



5. Discussion

5.1 Principal findings

Across 129 agentic educational AI studies, evaluation practice was systematically weaker than the pace and confidence of the literature would suggest. Three patterns recurred. First, evaluation was predominantly **output-oriented and proxy-based**: a majority of studies (56.6%) evaluated systems without human learners at all, and only 23.3% measured an actual learning outcome, with three-quarters relying solely on proxies such as output quality, perceptions, or process measures. Second, **internal validity and reproducibility were weakly established and weakly reported**: validated instruments (1.6% fully), novelty-effect control (2.3%), full protocol replicability (17.8%), and prompt release (50.4%) were all uncommon, and the modal study scored 2 of 5 on the MMAT. Third, the **reporting gaps concentrated in the appraisal-critical fields**: ethics status, instrument validity, and novelty control, precisely the fields that a reader most needs in order to judge whether a claim is credible.

The retrospective AEER scoring sharpened these into a single diagnosis. Studies disclosed most Essential items (mean 16.2 of 18) yet satisfied far fewer of the practice-bearing ones. The field, in other words, is closer to a *transparency* norm than to a *rigor* norm. Authors increasingly say what they did, but what they did frequently falls short of the designs and controls needed to support claims about learning. This gap between high disclosure and lower practice is the central contribution of the descriptive synthesis and the premise of the proposed standard.

5.2 Interpretation

The dominance of system-only and simulation evaluation is consistent with a field whose center of gravity is still engineering rather than education. Demonstrating that an agentic configuration produces better outputs than an ablation or a vanilla-LLM baseline answers a systems question, not an educational one. It establishes that a design choice changes model behavior, not that learners learn more. This echoes a long-standing concern in the learning sciences that instructional technology must be grounded in models of how learning actually occurs, not only in system capability (Koedinger, Corbett, & Perfetti, 2012). When fewer than one in four studies measures learning and only two in the corpus use a randomized design, the evidentiary basis for claims of educational benefit is, at present, thin.

The reproducibility findings are particularly consequential for *this* class of system. The behavior of an agentic educational system is defined largely by its prompts, orchestration, and tool configuration. When prompts are released in only half of studies and full protocols are reproducible in under a fifth, the intervention itself is under-specified, and independent verification, or even precise replication of a reported result, is often impossible. The near-universal “partial” replicability rating (82.2%) indicates that the problem is rarely total opacity but pervasive incompleteness: enough is reported to describe the system, not enough to rebuild and re-evaluate it. The wider machine-learning-based sciences have documented how incomplete reporting and methodological opacity drive reproducibility

failures (Kapoor & Narayanan, 2023); the agentic educational literature shows the same vulnerability at the point where its claims about learning are most consequential.

That the largest reporting gaps fall on instrument validity and ethics is telling. Both are low-cost to report and central to appraisal, yet both are frequently absent. The ethics gap is partly explained by the prevalence of simulation studies for which approval may not apply. Over the human-subjects subset, half reported approval, but the residual silence, and the near-absence of novelty-effect control, suggest that evaluation is often treated as a demonstration appended to a system paper rather than as a study designed to withstand scrutiny.

Crucially, the high disclosure rate reframes the problem as tractable. If near-complete disclosure is already common (71.3% of studies were non-silent on 16 or more of 18 Essential items), the remaining gaps are largely a matter of reporting convention, not of additional experimental burden. A standard that asks authors to state, for each item, whether and how they addressed it would close much of the gap without requiring anyone to run a larger study, and it would let readers distinguish “not done” from “not reported”, a distinction the current literature frequently obscures.

5.3 The AEER checklist as a response

The proposed Agentic Educational AI Evaluation and Reporting (AEER) checklist is designed against exactly the gaps observed. Each item maps to a measured deficiency: the agentic-profile and model-version items to the uneven characterization of systems; the design, comparator, and learning-outcome items to the proxy-and-simulation pattern; the prompt, code, and protocol items to the reproducibility shortfall; and the ethics, oversight, and safeguards items to the appraisal gaps. The checklist is deliberately a *disclosure* standard rather than a performance bar. It does not require any study to run an RCT or use a validated instrument, but it requires a study to state whether it did, so that an honest “not done” is compliant and silence is not. This design follows the finding that the field’s binding constraint is reporting, not effort, and it lowers the adoption cost. A checklist that demanded specific designs would be a stronger and more contestable claim, which was flagged as a separate question (Section 5.5).

I applied the checklist to the corpus to demonstrate that it is computable and discriminating, not to validate it; because its items were derived from the same data, the compliance figures are illustrative. Their value is to show that AEER separates studies (per-study disclosure ranged from 12 to 18 of 18) and that it localizes the field’s weaknesses to a small set of high-value items, which is what a usable reporting standard should do.

5.4 Implications

For **authors and reviewers**, the practical implication is immediate: a short, field-specific checklist applied at writing and review could materially raise the legibility of the evidence base at low cost, and would make the proxy-versus-learning distinction, which is the corpus’s dominant construct-validity risk, explicit rather than implied. For **the research community**, the findings argue for shifting evaluation from demonstrating capability

toward establishing educational effect, with human-learner designs, instructional comparators, and controls for novelty. For **practitioners and policymakers**, the central caution is that current claims about agentic educational systems rest substantially on proxy evidence and under-specified protocols. Any enthusiasm for such systems should be calibrated to that evidentiary base.

5.5 Limitations

The findings must be read against the review's design and scope, which is the more important caveat here given the subject. (1) **Open-accessibility scope.** The corpus comprises the 132 of 477 topically eligible records (28%) whose full text was openly accessible; because open-access status may correlate with venue, recency, and reporting norms, the findings characterize the openly accessible literature and should not be assumed to hold for paywalled venues. This limitation is of particular weight for a review whose findings concern reporting practice, since the very behavior under study could differ between open and paywalled venues. The direction of any such bias is not known. (2) **Single reviewer.** Screening and extraction were conducted by one reviewer; the blind human-AI agreement statistic ($\kappa = 0.65/0.74$) characterizes the advisory screening tool but does not substitute for independent human double-screening, and no formal extraction-reliability assessment was conducted. (3) **AI assistance.** A language model advised screening and drafted stated-fact extraction fields under an evidence-quotation constraint with human verification, and all judgment fields were human-coded; residual risk remains and the procedure is disclosed in full for scrutiny. (4) **Merged full-text eligibility and extraction and exclusion of "maybe" records** narrow the corpus at its boundary. (5) **Circularity of the compliance analysis.** AEER items were derived from the corpus gaps and then scored against the same corpus, so the compliance figures illustrate rather than validate the standard. (6) **English-only sources** and the search snapshot bound coverage. I note these not as caveats appended after the fact but as determinants of how far the present findings should be generalized: the patterns are, I believe, robust in direction, but their precise magnitudes pertain to the openly accessible corpus.

5.6 Future work

Whether the patterns reported here extend to paywalled venues is an open empirical question and a substantive one, since reporting norms may themselves differ by venue; a comparison across the open/paywalled boundary would test that directly. Subsequent work should also add a second independent reviewer with reported inter-rater and extraction reliability; validate the AEER checklist prospectively and independently of the corpus that generated it, ideally with a multi-rater reliability study and community consultation on item content and tiering; and extend the analysis to temporal trends as the literature matures. A worked application of AEER to a high- and a low-rigor study would further illustrate its use.

5.7 Conclusion

The agentic educational AI literature is expanding quickly, but its evaluation practice has not kept pace. Most systems are assessed without learners, few studies measure learning,

and the information needed to reproduce or appraise an evaluation is frequently incomplete. Encouragingly, studies already disclose most of what a reporting standard would ask. Thus, the gap is one of convention more than of effort. A focused reporting standard such as AEER, adopted at writing and review, offers a low-cost path from demonstration toward credible, cumulative evidence. Realizing that requires the field to treat evaluation not as a postscript to system-building but as the means by which its claims earn belief.

Author contributions. D.H. is the sole author and conducted all aspects of the work. CRediT roles: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Project administration; Software; Validation; Visualization; Writing – original draft; Writing – review & editing. All roles were performed by D.H.

Competing interests. The author declares no competing interests.

Ethics. This study is a systematic review of previously published research. It did not involve human participants, personal data, or animals, and no ethical approval was required.

Generative AI and AI-assisted technologies. Consistent with the procedures described in the Methods (screening and data extraction), this review used a large language model in two clearly bounded, non-binding roles: (i) as an advisory second screener at title/abstract screening, and (ii) as an evidence-locating aid during data extraction. In both roles the model's output was advisory only. The human author made every inclusion and exclusion decision, verified every AI-drafted stated-fact field against a required verbatim source quotation, and coded all judgment-based fields (design classification, appraisal, and the agentic-property determinations) without model input. The configurations, frozen prompts (with recorded hashes), and per-record provenance are reported in the Methods and released in the study repository; the specific model versions are identified in the Methods. No generative AI system is an author of this manuscript, and no generative AI system was used to generate its scholarly content, analysis, or conclusions.

Data and code availability. The review protocol, the verbatim database search strategies and search dates, and the record-identification and deduplication code are openly available in the supplementary repository at <https://github.com/HD013/agentive-educational-ai-review>. Raw bibliographic records exported from subscription databases (Scopus, Web of Science, IEEE Xplore, ACM Digital Library) are licensed and cannot be redistributed; the verbatim queries and search dates are provided so that users with database access can regenerate the corpus. Records from open sources (ERIC, arXiv, EdArXiv) are retrievable through the included public-API scripts. The screening configuration and provenance, the 41-field extraction codebook, the per-study extraction records, and a derived, identifier-level corpus with screening decisions are available in the study repository.

References

Cited references.

- Ba, S., Shi, X., Wu, S., et al. (2026). Artificial intelligence agents in computer-supported collaborative learning: A systematic literature review. *Computers and Education: Artificial Intelligence*, 10, 100579. <https://doi.org/10.1016/j.caeai.2026.100579>
- Chu, Z., Wang, S., Xie, J., Zhu, T., Yan, Y., Ye, J., Zhong, A., Hu, X., Liang, J., Yu, P. S., & Wen, Q. (2025). LLM Agents for Education: Advances and Applications. *arXiv:2503.11733*. <https://arxiv.org/abs/2503.11733> (Capability survey — excluded as a secondary study; also appeared in Findings of the ACL: EMNLP 2025.)
- Hong, Q. N., Fàbregues, S., Bartlett, G., Boardman, F., Cargo, M., Dagenais, P., Gagnon, M.-P., Griffiths, F., Nicolau, B., O’Cathain, A., Rousseau, M.-C., Vedel, I., & Pluye, P. (2018). The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. *Education for Information*, 34(4), 285–291. <https://doi.org/10.3233/EFI-180221>
- Jiang, Z., & Jiang, M. (2024). Beyond Answers: Large Language Model-Powered Tutoring System in Physics Education for Deep Learning and Precise Understanding. *arXiv:2406.10934*. <https://arxiv.org/abs/2406.10934> (Corpus RECID R5267; screened but not among the 132 extracted.)
- Kamalov, F., Santandreu Calonge, D., Smail, L., Azizov, D., Thadani, D. R., Kwong, T., & Atif, A. (2025). Evolution of AI in Education: Agentic Workflows. *arXiv:2504.20082*. <https://arxiv.org/abs/2504.20082>
- Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804. <https://doi.org/10.1016/j.patter.2023.100804>
- Koedinger, K. R., Corbett, A. T., & Perfetti, C. (2012). The Knowledge-Learning-Instruction framework: Bridging the science-practice chasm to enhance robust student learning. *Cognitive Science*, 36(5), 757–798. <https://doi.org/10.1111/j.1551-6709.2012.01245.x>
- Ouzzani, M., Hammady, H., Fedorowicz, Z., & Elmagarmid, A. (2016). Rayyan—a web and mobile app for systematic reviews. *Systematic Reviews*, 5, 210. <https://doi.org/10.1186/s13643-016-0384-4>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., et al. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372, n71. <https://doi.org/10.1136/bmj.n71>
- Stamper, J., Xiao, R., & Hou, X. (2024). Enhancing LLM-Based Feedback: Insights from Intelligent Tutoring Systems and the Learning Sciences. *arXiv:2405.04645*. <https://arxiv.org/abs/2405.04645>
- Wang, T., Zhan, Y., Lian, J., Hu, Z., Yuan, N. J., Zhang, Q., Xie, X., & Xiong, H. (2025). LLM-powered Multi-agent Framework for Goal-oriented Learning in Intelligent Tutoring System (GenMentor). In *Companion Proceedings of the ACM Web Conference 2025*. <https://doi.org/10.1145/3701716.3715244> (arXiv:2501.15749)

- Zhang, L., Deng, X., & Shadiev, R. (2026). A Meta-Review of Generative AI in Education: Synthesizing Findings from Systematic Reviews. *Journal of Educational Computing Research*, 64(4), 1068–1092. <https://doi.org/10.1177/07356331261419689>
-

Included studies

- Bakhtawar Ahtisham; Kirk Vanacore; Rene F. Kizilcec (2026). Optimizing LLM Annotation of Classroom Discourse through Multi-Agent Orchestration. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2603.13353> (R5938)
- Alshehri, A. (2025). Developing a multi-layer agent framework to enhance AI-generated educational questions for cybersecurity. *Journal of Umm Al-Qura University for Engineering and Architecture*, 16(4), 1045–1056. Springer Science and Business Media Deutschland GmbH. <https://doi.org/10.1007/s43995-025-00136-x> (R2182)
- An, S.; Li, Y.; Ma, Y.; Cheng, G.; Xu, G. (2024). Developing an LLM-Empowered Agent to Enhance Student Collaborative Learning Through Group Discussion. 32ND INTERNATIONAL CONFERENCE ON COMPUTERS IN EDUCATION CONFERENCE PROCEEDINGS, ICCE 2024, VOL II, 250–255. <https://doi.org/10.58459/icce.2024.4953> (R3912)
- Arana, J.M.P.; Carandang, K.A.M.; Casin, E.R.; Alis, C.; Tan, D.S.; Legara, E.F.; Monterola, C. (2025). Foundations of PEERS: Assessing LLM Role Performance in Educational Simulations. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 4, 908–918. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2025.acl-srw.66> (R2363)
- Reza Vatankhah Barenji; Nazila Salimi; Sina Khoshgoftar (2026). An LLM-Powered Assessment Retrieval-Augmented Generation (RAG) For Higher Education. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2601.06141> (R5634)
- Bendarkawi, J.; Ponce, A.; Mata, S.C.; Aliu, A.; Liu, Y.; Zhang, L.; Liaqat, A.; Rao, V.N.; Monroy-Hernández, A. (2025). ConversAR: Exploring Embodied LLM-Powered Group Conversations in Augmented Reality for Second Language Learners. *Conference on Human Factors in Computing Systems - Proceedings*. Association for Computing Machinery. <https://doi.org/10.1145/3706599.3720162> (R2105)
- Berrezueta-Guzman, S.; Wagner, S. (2026). Next-Gen orientation: supporting international students with generative AI NPCs in VR. *Frontiers in Computer Science*, 8, 1799323. Frontiers Media SA. <https://doi.org/10.3389/fcomp.2026.1799323> (R3558)
- Parth Bhalerao; Diola Dsouza; Ruiwen Guan; Oana Ignat (2026). Beyond Factual QA: Mentorship-Oriented Question Answering over Long-Form Multilingual Content. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2601.17173> (R5658)
- Madison Bochard; Tim Conser; Alyssa Duran; Lazaro Martull; Pu Tian; Yalong Wu (2025). LeafTutor: An AI Agent for Programming Assignment Tutoring. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2601.02375> (R5529)

- Jie Cao; Ha Nguyen; Selim Yavuz; Boran Yu; Shuguang Wang; Pavneet Kaur Bharaj; Dionne Cross Francis (2026). Developing Authentic Simulated Learners for Mathematics Teacher Learning: Insights from Three Approaches with Large Language Models. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2604.04361> (R5726)
- Chen, Y.; Ding, N.; Zheng, H.-T.; Liu, Z.; Sun, M.; Zhou, B. (2024). Empowering Private Tutoring by Chaining Large Language Models. International Conference on Information and Knowledge Management, Proceedings, 354–364. Association for Computing Machinery. <https://doi.org/10.1145/3627673.3679665> (R1479)
- Eason Chen; Isabel Wang; Nina Yuan; Sophia Judicke; Kayla Beigh; Xinyi Tang (2026). From Tool to Teammate: LLM Coding Agents as Collaborative Partners for Behavioral Labeling in Educational Dialogue Analysis. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2603.27440> (R5802)
- Cheng, C.; Zhao, G.; Huang, Z.; Zhuang, Y.; Pan, Z.; Liu, Q.; Li, X.; Chen, E. (2024). Towards Explainable Computerized Adaptive Testing with Large Language Model. EMNLP 2024 - 2024 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2024, 2655–2672. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2024.findings-emnlp.149> (R1714)
- Cohn, C.; Rayala, S.; Srivastava, N.; Fonteles, J.H.; Jain, S.; Luo, X.; Mereddy, D.; Mohammed, N.; Biswas, G. (2026). A Theory of Adaptive Scaffolding for LLM-Based Pedagogical Agents. Proceedings of the AAAI Conference on Artificial Intelligence, 40(3), 1757–1765. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v40i3.37154> (R3064)
- Clayton Cohn; Surya Rayala; Siyuan Guo; Hanchen David Wang; Naveeduddin Mohammed; Umesh Timalsina; Shruti Jain; Ryan Li; Angela Eeds; Menton Deweese; Pamela J. Osborn Popp; Rebekah Stanton; Shakeera Walker; Ashwin T S; Meiyi Ma; Gautam Biswas (2026). A Theory-Guided LLM Pedagogical Agent for STEM+C Scaffolding Without Over-Reliance. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2605.30539> (R5587)
- Copur-Gencturk, Y; Moreno, K; Chu, YC; Li, H; Tang, JL (2026). Automated coding of content and pedagogical content knowledge of mathematics using a multi-agent large language model. ZDM-MATHEMATICS EDUCATION.
<https://doi.org/10.1007/s11858-026-01796-2> (R4867)
- Daryanto, T.; Ding, X.; Wilhelm, L.T.; Stil, S.; Knutsen, K.M.; Rho, E.H. (2025). Conversate: Supporting Reflective Learning in Interview Practice Through Interactive Simulation and Dialogic Feedback. Proceedings of the ACM on Human-Computer Interaction, 9(1). Association for Computing Machinery. <https://doi.org/10.1145/3701188> (R2106)
- David, Jones; Ghosh, Shreya (2026). IntelliCode: A Multi-Agent LLM Tutoring System with Centralized Learner Modeling. Proceedings of the 19th Conference of the {E}uropean Chapter of the {A}ssociation for {C}omputational {L}inguistics (Volume 3: System Demonstrations), 129–138. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2026.eacl-demo.10> (R0156)

- de Fortuny, E.J. (2025). Simulating Market Equilibrium with Large Language Models. Proceedings of the Annual Hawaii International Conference on System Sciences, 4979–4986. IEEE Computer Society.
[https://www.scopus.com/pages/publications/105005147123?origin=resultslist\(R2834\)](https://www.scopus.com/pages/publications/105005147123?origin=resultslist(R2834))
- Dong, Z.; Dai, Z.; Lv, X.; Chen, J. (2026). Learning from Long-Term Engagement: Adaptive Tutoring Dialogue Planning for Personalized Education. Proceedings of the AAAI Conference on Artificial Intelligence, 40(1), 237–245. Association for the Advancement of Artificial Intelligence. [https://doi.org/10.1609/aaai.v40i1.36984\(R3497\)](https://doi.org/10.1609/aaai.v40i1.36984(R3497))
- Düsterbeck, LM; Grössler, M; Credidio, G; Bellmann, L; Riemann, LT (2026). Node-Sampling: adaptive multi-agent optimization in medical education. FRONTIERS IN COMPUTER SCIENCE, 8, 1758333.
[https://doi.org/10.3389/fcomp.2026.1758333\(R5036\)](https://doi.org/10.3389/fcomp.2026.1758333(R5036))
- Iizalaarab Elhaimeur; Nikos Chrisochoides (2026). From Prototype to Classroom: An Intelligent Tutoring System for Quantum Education. arXiv (Cornell University).
[https://doi.org/10.48550/arxiv.2604.24807\(R5795\)](https://doi.org/10.48550/arxiv.2604.24807(R5795))
- Iizalaarab Elhaimeur; Nikos Chrisochoides (2026). ITAS: A Multi-Agent Architecture for LLM-Based Intelligent Tutoring. arXiv (Cornell University).
[https://doi.org/10.48550/arxiv.2604.24808\(R5829\)](https://doi.org/10.48550/arxiv.2604.24808(R5829))
- Iizalaarab Elhaimeur; Nikos Chrisochoides (2026). Latency and Cost of Multi-Agent Intelligent Tutoring at Scale. arXiv (Cornell University).
[https://doi.org/10.48550/arxiv.2604.24110\(R5866\)](https://doi.org/10.48550/arxiv.2604.24110(R5866))
- Farhana, E.; Sarkar, S.; Knipper, R.; Dey, I.; Narayanan, H.; Puntambekar, S.; Karmaker, S. (2024). SimPal: Towards a Meta-Conversational Framework to Understand Teacher’s Instructional Goals for K-12 Physics. L@S 2024 - Proceedings of the 11th ACM Conference on Learning @ Scale, 461–465. Association for Computing Machinery, Inc. [https://doi.org/10.1145/3657604.3664695\(R1668\)](https://doi.org/10.1145/3657604.3664695(R1668))
- Fares Fawzi; Seyed Parsa Neshaei; Marta Knezevic; Tanya Nazaretsky; Tanja Käser (2026). REFINE: Real-world Exploration of Interactive Feedback and Student Behaviour. arXiv (Cornell University). [https://doi.org/10.48550/arxiv.2603.29142\(R5977\)](https://doi.org/10.48550/arxiv.2603.29142(R5977))
- Matthew Frazier; Kostadin Damevski (2026). Towards Personalizing Secure Programming Education with LLM-Injected Vulnerabilities. arXiv (Cornell University). [https://doi.org/10.48550/arxiv.2604.13955\(R6084\)](https://doi.org/10.48550/arxiv.2604.13955(R6084))
- Quanzhi Fu; Qiyu Wu; Dan Williams (2025). Mining the Gold: Student-AI Chat Logs as Rich Sources for Automated Knowledge Gap Detection. arXiv (Cornell University). [https://doi.org/10.48550/arxiv.2512.22404\(R5538\)](https://doi.org/10.48550/arxiv.2512.22404(R5538))
- Gou, P.; Teng, T.; Wang, L. (2025). Research on the Application of Metacognitive Teaching Scaffolding Empowered by GenAI in Deep Learning - - Taking the vocational school course “Fundamentals and Skills of Electronic Technology” as an example. Proceedings of 2025 6th International Conference on Education, Knowledge and Information Management, ICEKIM 2025, 430–434. Association for Computing Machinery, Inc. [https://doi.org/10.1145/3756580.3756651\(R2790\)](https://doi.org/10.1145/3756580.3756651(R2790))

- Shuchen Guo; Ehsan Latif; Yifan Zhou; Xuan Huang; Xiaoming Zhai (2024). Using Generative AI and Multi-Agents to Provide Automatic Feedback. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2411.07407> (R5459)
- Hao, Y.; Holmes, J.; Waddle, M.R.; Davis, B.J.; Yu, N.Y.; Vickers, K.S.; Preston, H.; Margolin, D.; Löckenhoff, C.E.; Vashistha, A.; Kalantari, S.; Ghassemi, M.; Liu, W. (2025). Personalizing prostate cancer education for patients using an EHR-Integrated LLM agent. npj Digital Medicine, 8(1). Nature Research. <https://doi.org/10.1038/s41746-025-02166-0> (R2727)
- Hou, X.; Liu, Z.; Wang, G.; Shi, N. (2026). Bridging awareness and behavior: decoding implicit metacognitive behaviors in AI-assisted programming via fine-grained log analysis. Frontiers in Psychology, 17, 1790196. Frontiers Media SA. <https://doi.org/10.3389/fpsyg.2026.1790196> (R3201)
- Hu, Z.; Paprotskyi, M.-M.; Vysotska, V.; Chyrun, L.; Ushenko, Y.; Uhryn, D. (2025). Local Agentic RAG-Based Information System Development for Intelligent Analysis of GitHub Code Repositories in Computer Science Education. International Journal of Modern Education and Computer Science, 17(5), 109–145. Modern Education and Computer Science Press. <https://doi.org/10.5815/ijmecs.2025.05.07> (R2609)
- Fareya Ikram; Nischal Ashok Kumar; Junyang Lu; Hunter McNichols; Candace Walkington; Neil T. Heffernan; Andrew S. Lan (2026). A Multi-Agent Approach to Validate and Refine LLM-Generated Personalized Math Problems. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2604.05160> (R5579)
- Jain, R; Rengarajan, S (2025). Emergent Wisdom at BEA 2025 Shared Task: From Lexical Understanding to Reflective Reasoning for Pedagogical Ability Assessment. PROCEEDINGS OF THE 20TH WORKSHOP ON INNOVATIVE USE OF NLP FOR BUILDING EDUCATIONAL APPLICATIONS, BEA 2025, 1108–1120. <https://doi.org/10.18653/v1/2025.bea-1.86> (R4369)
- Jang, J.; Moon, A.; Jung, M.; Kim, Y.; Lee, S.J. (2025). LLM Agents at the Roundtable: A Multi-Perspective and Dialectical Reasoning Framework for Essay Scoring. EMNLP 2025 - 2025 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2025, 19674-19687. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2025.findings-emnlp.1072> (R2528)
- Jiang, Y.-H.; Tang, K.; Chen, Z.-W.; Wei, Y.; Liu, T.-Y.; Wu, J. (2025). MAS-KCL: knowledge component graph structure learning with large language model-based agentic workflow. Visual Computer, 41(9), 6453–6464. Springer Science and Business Media Deutschland GmbH. <https://doi.org/10.1007/s00371-025-03946-1> (R2620)
- W.M. Wang L.H. Jiang; Shiyuan Wang; Hiba Eltigani; Rukhshan Haroon; Abdullah Bin Faisal; Fahad Dogar (2025). AdvisingWise: Supporting Academic Advising in Higher Education Settings Through a Human-in-the-Loop Multi-Agent Framework. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2511.05706> (R5482)
- Jin, S.; Wang, H.; Gao, Z.; Yang, Y.; Chunjia, B.; Wang, C. (2025). Evolution in Simulation: AI-Agent School with Dual Memory for High-Fidelity Educational Dynamics. EMNLP 2025 - 2025 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2025, 5843–5857. Association for

Computational Linguistics (ACL).

<https://doi.org/10.18653/v1/2025.findings-emnlp.312> (R2298)

- Jin, H.; Yoo, M.; Park, J.; Lee, Y.; Wang, X.; Kim, J. (2025). TeachTune: Reviewing Pedagogical Agents Against Diverse Student Profiles with Simulated Students. Conference on Human Factors in Computing Systems - Proceedings. Association for Computing Machinery. <https://doi.org/10.1145/3706598.3714054> (R2881)
- Shi Jinxin; Jiabao Zhao; Yilei, Wang; Xingjiao Wu; Jiawen Li; Liang He (2023). CGMI: Configurable General Multi-Agent Interaction Framework. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2308.12503> (R5147)
- Jordán, J.; Yin, X.; Fabros, M.; Ranade, G.; Norouzi, N. (2026). MAGIC: Multi-Agent Argumentation and Grammar Integrated Critiquer. Proceedings of the AAAI Conference on Artificial Intelligence, 40(47), 40599-40607. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v40i47.41506> (R3515)
- Nizam Kadir (2026). From Untamed Black Box to Interpretable Pedagogical Orchestration: The Ensemble of Specialized LLMs Architecture for Adaptive Tutoring. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2603.23990> (R5803)
- Rafal Kocielnik; J. Everett Knudsen; Steven Y. Cen; Jasmine Lin; Cherine H. Yang; Atharva Deo; Ujjwal Pasupulety; Peter Wager; Anima Anandkumar; Andrew J. Hung (2026). A Multi-Agent LLM Framework for Rating the Quality of Surgical Feedback. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2605.25440> (R5581)
- Kricheli, J.S.; Nye, B.; Auerbach, D.; Shiel, A.; Carr, K.; Alison, K.; Swartout, W. (2026). Evaluating Personalized Content Using Large Language Models. Proceedings of the International Florida Artificial Intelligence Research Society Conference, FLAIRS, 39(1), 1–9. Florida Online Journals, University of Florida. <https://doi.org/10.32473/flairs.39.1.141858> (R3329)
- Harsh Kumar; Zi Kang; Mu; Jonathan Vincentius; Ashton Anderson (2026). Beyond the AI Tutor: Social Learning with LLM Agents. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2604.02677> (R5665)
- Lee, J.; Park, K.; Park, J. (2024). VISTA: Visual Integrated System for Tailored Automation in Math Problem Generation Using LLM. Proceedings of Machine Learning Research, 264, 136–156. ML Research Press. <https://www.scopus.com/pages/publications/85219555730?origin=resultlist> (R1740)
- Lee, J.; Chen, S.-S.; Liang, P.P. (2025). Interactive Sketchpad: A Multimodal Tutoring System for Collaborative, Visual Problem-Solving. Conference on Human Factors in Computing Systems - Proceedings. Association for Computing Machinery. <https://doi.org/10.1145/3706599.3719790> (R2510)
- Li, Y.; Liu, J.; Hu, Y.; Liu, C.; Zhang, Y.; Yin, L.; Dong, W. (2025). Beyond Test Cases: Multi-Agent Collaboration for Detecting Defects in Full-Score Code Implementations. Proceedings of the International Conference on Software Engineering and Knowledge Engineering, SEKE, 124–129. Knowledge Systems Institute Graduate School. <https://doi.org/10.18293/seke2025-053> (R2016)

- Li, H.; Xu, T.; Chang, E.; Wen, Q. (2025). Knowledge Tagging with Large Language Model Based Multi-Agent System. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(28), 28775-28782. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v39i28.35141> (R2523)
- Li, G.; Chen, L.; Tang, C.; Švábenský, V.; Deguchi, D.; Yamashita, T.; Shimada, A. (2025). Single-agent vs. Multi-agent LLM Strategies for Automated Student Reflection Assessment. *Lecture Notes in Computer Science*, 15874 LNCS, 300–311. Springer Science and Business Media Deutschland GmbH. https://doi.org/10.1007/978-981-96-8186-0_24 (R2837)
- Shuang Li; Haiyang Xin; Yimeng Sun; Qiannan Niu; Lingyun Huang; Gaowei Chen; Ching Sing Chai (2026). MIRACLE_Multi-Agent Intelligent Regulation to Advance Collaborative Learning Environment. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2605.12923> (R5885)
- Yuejia Li; Ke He; Junheng Li; Shutong Chen; Jingkang Xia; Zhiyue Su; Junchi Zhang; Mang Ye (2026). See Before You Code: Learning Visual Priors for Spatially Aware Educational Animation Generation. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2605.15585> (R6031)
- Belle Li; Lily Tan; Wei Zakharov; Qiang Qiu; Colby Ben Acton (2026). Tutor, Not Solver: Designing a Guardrailed AI Assistant for Learning in Higher Education: A Design Case of PeteChat. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2606.09845> (R6094)
- Li, X.; Li, T.; Yan, L.; Li, Y.; Zhao, L.; Raković, M.; Molenaar, I.; Gašević, D.; Fan, Y. (2026). FLoRA: An advanced AI-powered engine to facilitate hybrid human-AI regulated learning. *Computers and Education*, 243, 105527. Elsevier. <https://doi.org/10.1016/j.compedu.2025.105527> (R3350)
- Yuxin Liu; Zeqing Song; Jiong Lou; Chentao Wu; Jie Li (2025). AgentTutor: Empowering Personalized Learning with Multi-Turn Interactive Teaching in Intelligent Education Systems. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2601.04219> (R5485)
- Tao Liu; Ye Lu; Ruohua Zhang; Siyu Song; Wentao Liu; Aimin Zhou; Hao Hao (2026). Elmes*: Automated Construction of Fine-Grained Evaluation Rubrics for Large Language Models in Long-Tail Educational Scenarios. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2606.06546> (R5751)
- Jiarui Liu; Terry Jingchen Zhang; Ryan Faulkner; X. Angelo Huang; Vilém Zouhar; Dominik Glandorf; Isabel Dahlgren; Van Q. Truong; Rishit Dagli; Yuen Chen; Felix Leeb; Punya Syon Pandey; Yves Bicker; Suvajit Majumder; Wenyuan Jiang; Zeju Qiu; Sankalan Pal Chowdhury; Bernhard Schölkopf; Mona Diab; Zhijing Jin (2026). PaperMentor: A Human-Centered Multi-Agent Writing Tutor for AI Research Papers on Overleaf. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2606.08857> (R5950)
- Yun Liang Long; Yu Zhang (2024). Enhanced Classroom Dialogue Sequences Analysis with a Hybrid AI Agent: Merging Expert Rule-Base with Large Language Models. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2411.08418> (R5307)

- Yiping Ma; Shiyu Hu; Xuchen Li; Yipei Wang; Yuqing Chen; Shiqing Liu; Kang Hao Cheong (2024). When LLMs Learn to be Students: The SOEI Framework for Modeling and Evaluating Virtual Student Agents in Educational Interaction. arXiv. <https://doi.org/10.48550/arXiv.2410.15701> (R5468)
- Victor De Marez; Jens Van Nooten; Luna De Bruyne; Walter Daelemans (2025). An Agentic AI Framework for Training General Practitioner Student Skills. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2512.18440> (R5489)
- Jakub MASŁOWSKI; Jarosław A. Chudziak (2026). Heterogeneous Debate Engine: Identity-Grounded Cognitive Architecture for Resilient LLM-Based Ethical Tutoring. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2603.27404> (R5820)
- Menze, D; Radovic, S; Seidel, N (2026). AI-assisted MCQ creation increases item-writing flaws through automation bias: evaluation of a workflow of multiple AI agents with teachers. FRONTIERS IN COMPUTER SCIENCE, 8, 1831250. <https://doi.org/10.3389/fcomp.2026.1831250> (R4834)
- Movidi, S., & Russell, S. (2025). The Personalization Paradox: Semantic Loss vs. Reasoning Gains in Agentic AI Q&A. arXiv:2512.04343. (R5562)
- Nguyen, L.; Nguyen, D.; Bui, Q.; Phan, D.; Le, D.; Nguyen, K.; Vo, Q.; Vo, K.; Duong, N.; Dinh, A.; Trinh, T.; Phan, C.; Nguyen, A.; Nguyen, T.; Le, D.; Dang, V.; Quan, T. (2026). Catching the First Light of Tomorrow: A Hackathon-Based Framework for Introducing High School Students to AI Agents. Proceedings of the AAAI Conference on Artificial Intelligence, 40(47), 40779-40787. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v40i47.41527> (R3212)
- Matthew Nyaaba; Patrick Kyeremeh; Macharious Nabang; Bismark Nyaaba Akanzire; Acquah, Sakina; Titty, Cyril Ababio; Asare, Kotor; Jerry Etornam Kudaya (2025). GenAITEd Ghana: A First-of-Its-Kind Context-Aware and Curriculum-Aligned Conversational AI Agent for Teacher Education. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2601.06093> (R5516)
- Odeajo, Israel Olanrewaju; Jimmy, Edifon Emmanuel (2026). EduNaija AI Tutor: A Multi-Agent Retrieval-Augmented Generation System for Nigerian Curriculum Education. Proceedings of the 7th Workshop on {A}frican Natural Language Processing ({A}frica{NLP} 2026), 113–115. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2026.africanlp-main.11> (R0152)
- Pedroche-Santoveña, I., & Feliz-Murias, T. (2025). Critical Media Literacy With and Within Generative AI: Design-Based Research on #PinchaLaBurbuja platform. VISUAL Review, 17(6), 209–232. <https://doi.org/10.62161/revvisual.v17.5948> (R2046)
- Pei, J.; Viola, I.; Huang, H.; Wang, J.; Ahsan, M.; Ye, F.; Jiang, Y.; Sai, Y.; Wang, D.; Chen, Z.; Ren, P.; Cesar, P. (2024). Autonomous Workflow for Multimodal Fine-Grained Training Assistants Towards Mixed Reality. Proceedings of the Annual Meeting of the Association for Computational Linguistics, 4051–4066. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2024.findings-acl.240> (R1395)
- Pei, J.; Ye, F.; Sun, X.; Deng, W.; Hindriks, K.; Wang, J. (2025). Conversational Education at Scale: A Multi-LLM Agent Workflow for Procedural Learning and

Pedagogic Quality Assessment. EMNLP 2025 - 2025 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2025, 2984–2997. Association for Computational Linguistics (ACL).
<https://doi.org/10.18653/v1/2025.findings-emnlp.161> (R2109)

- Peng, Xian; Yuan, Pan; Li, Dong; Cheng, Junlong; Fang, Qin; Liu, Zhi (2025). KELE: A Multi-Agent Framework for Structured Socratic Teaching with Large Language Models. Findings of the Association for Computational Linguistics: EMNLP 2025, 16342–16362. Association for Computational Linguistics.
<https://doi.org/10.18653/v1/2025.findings-emnlp.888> (R0092)
- Likai Peng; Shihui Feng (2026). Single-agent vs. Multi-agents for Automated Video Analysis of On-Screen Collaborative Learning Behaviors. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2604.03631> (R6037)
- Pérez-Ortiz, J.A.; Esplà-Gomis, M.; Sánchez-Cartagena, V.M.; Sánchez-Martínez, F.; Chernysh, R.; Mora-Rodríguez, G.; Berezhnoy, L. (2024). A Conversational Intelligent Tutoring System for Improving English Proficiency of Non-Native Speakers via Debriefing of Online Meeting Transcriptions. Proceedings of the 13th Workshop on Natural Language Processing for Computer Assisted Language Learning, NLP4CALL 2024, 187–198. Association for Computational Linguistics (ACL).
<https://www.scopus.com/pages/publications/105000124173?origin=resultslist> (R1329)
- David Power (2026). Can Large Language Models Generate Role-Consistent Clinical Dialogue for Education? A Multi-agent Approach. EdArXiv. https://osf.io/etv6d_v1 (R0655)
- Bahar Radmehr; Adish Singla; Tanja Käser (2024). Towards Generalizable Agents in Text-Based Educational Environments: A Study of Integrating RL with LLMs. International Educational Data Mining Society. <http://eric.ed.gov/?id=ED675590> (R0488)
- Rumble, T.; Zarrin, J.; Lovell, P.G.; Falconer, R. (2026). Learning Engagement Assistant (LEA): A Multi-Agent AI Framework for Adaptive and Personalized Learning with Simulated Student Agents. International Conference on Agents and Artificial Intelligence, 1, 512–523. Science and Technology Publications, Lda.
<https://doi.org/10.5220/0014418900004052> (R3493)
- Saisab Sadhu; Ashim Dhor (2025). Hierarchical Pedagogical Oversight: A Multi-Agent Adversarial Framework for Reliable AI Tutoring. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2512.22496> (R5520)
- Sanyal, Debdeep; Maiti, Agniva; Maharana, Umakanta; Kumar, Dhruv; Mali, Ankur; Giles, C. Lee; Mandal, Murari (2025). Investigating Pedagogical Teacher and Student LLM Agents: Genetic Adaptation Meets Retrieval-Augmented Generation Across Learning Styles. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, 13348–13389. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.emnlp-main.675> (R0090)
- Nicy Scaria; Silvester John Joseph Kennedy; Deepak Subramani (2026). Learning in Blocks: A Multi Agent Debate Assisted Personalized Adaptive Learning Framework for Language Learning. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2604.22770> (R5869)

- Schmucker, R.; Xia, M.; Azaria, A.; Mitchell, T. (2024). Ruffle & Riley: Insights from Designing and Evaluating a Large Language Model-Based Conversational Tutoring System. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 75–90. Springer Science and Business Media Deutschland GmbH.
https://doi.org/10.1007/978-3-031-64302-6_6 (R1653)
- Schmucker, R., Xia, M., Azaria, A., & Mitchell, T. (2023). Ruffle&Riley: Towards the Automated Induction of Conversational Tutoring Systems. *arXiv:2310.01420*. (R5205)
- Scholz, N.; Nguyen, M.H.; Singla, A.; Nagashima, T. (2025). Partnering with AI: A Pedagogical Feedback System for LLM Integration Into Programming Education. *Lecture Notes in Computer Science*, 243–248. Springer.
https://doi.org/10.1007/978-3-032-03873-9_31 (R3574)
- Schorcht, S.; Peters, F.; Kriegel, J. (2025). Communicative AI Agents in Mathematical Task Design: A Qualitative Study of GPT Network Acting as a Multi-professional Team. *Digital Experiences in Mathematics Education*, 11(1), 77–113. Springer International Publishing. <https://doi.org/10.1007/s40751-024-00161-w> (R2090)
- Dmitriy Sergeev (2025). Multimodal Talent Discovery in Children Using Calibrated Baselines. *EdArXiv*. <https://doi.org/10.5281/zenodo.17941256> (R0652)
- Shi, Y.; Liang, R.; Xu, Y. (2025). EducationQ: Evaluating LLMs' Teaching Capabilities Through Multi-Agent Dialogue Framework. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, 1, 32799–32828. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2025.acl-long.1576> (R2230)
- Sit, CYE; Yang, Y; Yeung, WK; Kong, SC (2024). Developing a LLMs-Driven System Based on Human-AI Progressive Code Generation Framework to Assist Mathematics Learning. *32ND INTERNATIONAL CONFERENCE ON COMPUTERS IN EDUCATION CONFERENCE PROCEEDINGS, ICCE 2024, VOL I*, 83–88.
<https://doi.org/10.58459/icce.2024.4815> (R3911)
- Jiajia Song; Zhihan Guo; Jionghao Lin (2026). Simulating Novice Students Using Machine Unlearning and Relearning in Large Language Models. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2603.26142> (R6035)
- Damji Stratton (2026). OBI-WAN: An AI Advising System for Human–AI Collaborative Education-to-Career Decisions. *EdArXiv*. https://osf.io/rp82m_v1 (R0658)
- Sun, F.; Li, L.; Meng, S.; Teng, X.; Payne, T.R.; Craig, P. (2025). Integrating emotional intelligence, memory architecture, and gestures to achieve empathetic humanoid robot interaction in an educational setting. *Frontiers in Robotics and AI*, 12, 1635419. Frontiers Media SA. <https://doi.org/10.3389/frobt.2025.1635419> (R2492)
- Sushri, S.; Dass, R.K.; Basappa, R.; Lu, H.; Goel, A.K. (2024). Combining Cognitive and Generative AI for Self-explanation in Interactive AI Agents. *CEUR Workshop Proceedings*, 3840. CEUR-WS.
<https://www.scopus.com/pages/publications/85210883719?origin=resultlist> (R1426)

- Tan, A.; Dorneich, M.C.; Cotos, E. (2025). Developing and evaluating the fidelity of virtual reality-artificial intelligence (VR-AI) environment for situated learning. *Frontiers in Virtual Reality*, 6, 1587768. Frontiers Media SA. <https://doi.org/10.3389/frvir.2025.1587768> (R2183)
- Tang, D.; Ding, R.; He, M.; Wang, Y.; Cheng, K. (2026). Intelligent teaching design assistant for primary mathematics: A large language model-driven framework with retrieval-augmented generation and problem-chain pedagogy. *International Electronic Journal of Mathematics Education*, 21(1). Modestum LTD. <https://doi.org/10.29333/iejme/17447> (R3448)
- Yinghao Tang; Yupeng Xie; Feng, Yingchaojie; Tian Lan; Wei Chen (23863) (2026). Demonstrating ViviDoc: Generating Interactive Documents through Human-Agent Collaboration. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2603.01912> (R5722)
- Tao, M.; Tang, B.; Ma, M.; Zhang, Y.; Li, H.; Wen, F.; Ma, H.; Yang, J. (2025). ALOHA: Empowering Multilingual Agent for University Orientation with Hierarchical Retrieval. *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies: Long Papers, NAACL-HLT 2025*, 6, 383–392. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2025.naacl-demo.32> (R1873)
- Aditya Thole; Anmol Agrawal; Arnav Ramamoorthy; Dhruv Kumar (2026). PhysicsSolutionAgent: Towards Multimodal Explanations for Numerical Physics Problem Solving. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2601.13453> (R5953)
- Bismack Tokoli; Luis G. Jaimes; Ayesha S. Dina (2026). ALIGNAgent: Adaptive Learner Intelligence for Gap Identification and Next-step guidance. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2601.15551> (R5594)
- Tran-Le, B.-T.; Thomas, P.; Stiffler, N.M.; Nguyen, T.N. (2026). How Does LLM-powered Coding Assistance Shape Incidental Learning? Exploring Cognitive Forcing Strategies in Programming Education. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(48), 40880-40888. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v40i48.42121> (R3406)
- Tripathi, H.; Crowell, C.; Newlin, K.; Neupane, S.; Rahimi, S.; Keith, J. (2026). ACE-TA: An Agentic Teaching Assistant for Grounded Q&A, Quiz Generation, and Code Tutoring. *Proceedings of the International Florida Artificial Intelligence Research Society Conference, FLAIRS*, 39(1). Florida Online Journals, University of Florida. <https://www.scopus.com/pages/publications/105040226128?origin=resultslist> (R3084)
- Candace Walkington; Theodora Beauchamp; Fareya Ikram; Merve Koçyiğit Gürbüz; Fangli Xia; Margan Lee; Andrew Lan (2026). Mathematics Teachers Interactions with a Multi-Agent System for Personalized Problem Generation. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2604.12066> (R5898)
- Wang, J.; Dai, Y.; Zhang, Y.; Ma, Z.; Li, W.; Chai, J. (2025). Training Turn-by-Turn Verifiers for Dialogue Tutoring Agents: The Curious Case of LLMs as Your Coding

Tutors. Proceedings of the Annual Meeting of the Association for Computational Linguistics, 12416-12436. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2025.findings-acl.642> (R2949)

- Wang, Y.; Ding, Z.; Wu, X.; Sun, S.; Liu, N.; Zhai, X. (2026). AutoSCORE: Enhancing Automated Scoring with Multi-Agent Large Language Models via Structured Component Recognition. Proceedings of the AAAI Conference on Artificial Intelligence, 40(48), 40898-40906. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v40i48.42123> (R3180)
- Baiqiang Wang; Yan Bai; Juan Li (2026). CyberJustice Tutor: An Agentic AI Framework for Cybersecurity Learning via Think-Plan-Act Reasoning and Pedagogical Scaffolding. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2603.18470> (R5712)
- Tianfu Wang; Max Xiong; Jianxun Lian; Hongyuan Zhu; Zhengyu Hu; Yuxuan Lei; Linxiao Gong; Xiaofang Li; Peiting Tsai; Nicholas Jing Yuan; Qi Zhang (2026). SocialCoach: Personalized Social Skill Learning with RL-based Agentic Tutoring and Practice. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2606.04155> (R6050)
- Yuang Wei; Ruijia Li; Bo Jiang (2026). SLOW: Strategic Logical-inference Open Workspace for Cognitive Adaptation in AI Tutoring. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2603.28062> (R6014)
- Wu, Zhenyu; Li, Jian; Huang, Hua (2025). MAGMA-Edu: Multi-Agent Generative Multimodal Framework for Text-Diagram Educational Question Generation. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2511.18714> (R5532)
- Wu, Yang; Yao, Rujing; Zhang, Tong; Shi, Yufei; Jiang, Zhuoren; Li, Zhushan; Liu, Xiaozhong (2025). Teaching According to Students' Aptitude: Personalized Mathematics Tutoring via Persona-, Memory-, and Forgetting-Aware LLMs. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2511.15163> (R5560)
- Han Wua; Lishan Zhang; Chunming Lu (2026). A Simulation-Based Method for Testing Collaborative Learning Scaffolds Using LLM-Based Multi-Agent Systems. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2604.11161> (R5585)
- Xi, Haoran; Shao, Minghao; Milner, Kimberly; Putrevu, Venkata Sai Charan; Rani, Nanda; Udeshi, Meet; Krishnamurthy, Prashanth; Dolan-Gavitt, Brendan; Garg, Siddharth; Shukla, Sandeep Kumar; Khorrami, Farshad; Hillel-Tuch, Alon; Shafique, Muhammad; Karri, Ramesh (2026). AI In Cybersecurity Education – Scalable Agentic CTF Design Principles and Educational Outcomes. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2603.21551> (R5590)
- Xie, E.; Waterfield, D.; Kennedy, M.; Zhang, A. (2026). SlideBot: A Multi-Agent Framework for Generating Informative, Reliable, Multi-Modal Presentations. Proceedings of the AAAI Conference on Artificial Intelligence, 40(48), 40907-40915. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v40i48.42124> (R3655)
- Xu, Zhiyu; Yan, Weilong; Shi, Yufei; Meng, Xin; He, Tao; Zhuang, Huiping; Li, Ming; Fan, Hehe (2025). SciEducator: Scientific Video Understanding and Educating via

Deming-Cycle Multi-Agent System. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2511.17943> (R5552)

- Haoxin Xu; Changyong Qi; Tong Liu; Bohao Zhang; Anna He; Bingqian Jiang; Longwei Zheng; Xiaoqing Gu (2026). Multi-Agent Learning Path Planning via LLMs. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2601.17346> (R5917)
- Lingyong Yan; Jiulong Wu; Dong Xie; Weixian Shi; Deguo Xia; Jizhou Huang (2026). Beyond End-to-End Video Models: An LLM-Based Multi-Agent System for Educational Video Generation. arXiv (Cornell University).
<https://doi.org/10.1145/3770855.3818323> (R5657)
- Qian Yang; Jessie Jia; Elaine Tsai; Amy X. Li; Nader Akoury; Natalya N. Bazarova (2026). Attention: What Prevents Young Adults from Speaking Up Against Cyberbullying in an LLM-Powered Social Media Simulation. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2605.03287> (R5642)
- Yao, Z.; Parashar, A.; Zhou, H.; Jang, W.S.; Ouyang, F.; Yang, Z.; Yu, H. (2025). MCQG-SRefine: Multiple Choice Question Generation and Evaluation with Iterative Self-Critique, Correction, and Comparison Feedback. Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies: Long Papers, NAACL-HLT 2025, 1, 10728-10777. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2025.naacl-long.538> (R2622)
- Yao, H.; Xu, W.; Turnau, J.; Kellam, N.; Wei, H. (2026). Instructional Agents: Reducing Teaching Faculty Workload through Multi-Agent Instructional Design. EACL 2026 - 19th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference, Vol. 1 - (Long Papers), 1, 4087-4109. Association for Computational Linguistics (ACL).
<https://doi.org/10.18653/v1/2026.eacl-long.191> (R3432)
- Tahreem Yasir; Wenbo Li; Sam Gilson; Sutapa Dey Tithi; Xiaoyi Tian; Tiffany Barnes (2026). Confirming Correct, Missing the Rest: LLM Tutoring Agents Struggle Where Feedback Matters Most. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2605.16207> (R5704)
- Yulei Ye; Wenhao Li; Zhong Wen; Yunshu Huang; Yichen Hu; Zifan Wei; Yige Wang; Xinyu Xie; Haoxuan Yang; Yanjun Huang; Ruijia Li; Hong Qian; Dr Yu Song; Bo Jiang; Bingdong Li; Lijun Li; Bo Zhang; Pinlong Cai; Xingcheng Xu; Shuangye Chen; Xia Hu; Liang He; Aimin Zhou; Jingjing Qu; Jing Shao; Xiangfeng Wang (2026). AgentSchool: An LLM-Powered Multi-Agent Simulation for Education. arXiv (Cornell University).
<https://doi.org/10.48550/arxiv.2605.30144> (R5615)
- Jifan Yu; Zheyuan Zhang; Daniel Zhang-Li; Shangqing Tu; Zhanxin Hao; Rui Li; Haoxuan Li; Yuanchun Wang; Hanming Li; Linlu Gong; Jie Cao; Jiayin Lin; Jinchang Zhou; Fei Qin; Haohua Wang; Jianxiao Jiang; Lijun Deng; Zhan, Yisi; Chaojun Xiao; Dai, Xusheng; Yan Xuan; Nianyi Lin; Nan Zhang; R. Ni; Yang Dang; Lei Hou; Yu Zhang; Xu Han; Manli Li; Juanzi Li; Zhiyuan Liu; Huiqin Liu; Maosong Sun (2024). From MOOC to MAIC: Reshaping Online Teaching and Learning through LLM-driven Agents. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.2409.03512> (R5321)

- Yu, H.; Zhou, J.; Li, L.; Chen, S.; Gallifant, J.; Shi, A.; Sun, J.; Li, X.; He, J.; Hua, W.; Jin, M.; Chen, G.; Zhou, Y.; Li, Z.; Gupte, T.; Chen, M.-L.; Azizi, Z.; Dou, Q.; Yan, B.P.; Xing, Y.; Zhang, Y.; Assimes, T.L.; Bitterman, D.S.; Ma, X.; Lu, L.; Fan, L. (2025). Simulated patient systems powered by large language model-based AI agents offer potential for transforming medical education. *Communications Medicine*, 6, 27. *Nature Research*. <https://doi.org/10.1038/s43856-025-01283-x> (R3650)
- Yusuf, H.; Money, A.; Daylamani-Zad, D. (2025). Towards reducing teacher burden in Performance-Based assessments using aivalue: an emotionally intelligent LLM-Augmented pedagogical AI conversational agent. *Education and Information Technologies*, 30(17), 24649-24693. Springer. <https://doi.org/10.1007/s10639-025-13755-7> (R2947)
- Zamojska, M.; Chudziak, J.A. (2025). TACLA: An LLM-Based Multi-Agent Tool for Transactional Analysis Training in Education. *Proceedings - International Conference on Tools with Artificial Intelligence, ICTAI*, 313–320. IEEE Computer Society. <https://doi.org/10.1109/ictai66417.2025.00048> (R2867)
- Zha, S.; Liu, Y.; Zheng, C.; Xu, J.; Yu, F.; Gong, J.; Xu, Y. (2025). Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process. *Conference on Human Factors in Computing Systems - Proceedings*. Association for Computing Machinery. <https://doi.org/10.1145/3706598.3713952> (R2645)
- Zhang, Z.; Zhang-Li, D.; Yu, J.; Gong, L.; Zhou, J.; Hao, Z.; Jiang, J.; Cao, J.; Liu, H.; Liu, Z.; Hou, L.; Li, J. (2025). Simulating Classroom Education with LLM-Empowered Agents. *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies: Long Papers, NAACL-HLT 2025*, 1, 10364-10379. Association for Computational Linguistics (ACL). <https://doi.org/10.18653/v1/2025.naacl-long.520> (R2832)
- Zhang, C.; Luo, X. (2026). Scaling Equitable Reflection Assessment in Education via Large Language Models and Role-Based Feedback Agents. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(46), 39594-39602. Association for the Advancement of Artificial Intelligence. <https://doi.org/10.1609/aaai.v40i46.41311> (R3638)
- Zhang-Li, D.; Zhang, Z.; Yu, J.; Lim, J.J.Y.; Tu, S.; Gong, L.; Wang, H.; Liu, Z.; Liu, H.; Hou, L.; Li, J. (2025). Awakening the Slides: A Tuning-free and Knowledge-regulated AI Tutoring System via Language Model Coordination. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1, 2872–2883. Association for Computing Machinery. <https://doi.org/10.1145/3690624.3709423> (R1998)
- Zhang, B., Liu, X., Wang, Y., Zhou, L., Xie, Q., & Wang, B. (2025). Human or LLM as Standardized Patients? A Comparative Study for Medical Education. *arXiv:2511.14783*. (R5522)
- Bingxi Zhao; Jiahao Zhang; Xubin Ren; Zirui Guo; Tianzhe Chu; Yi Ma; Chao Huang (2026). DeepTutor: Towards Agentic Personalized Tutoring. *arXiv* (Cornell University). <https://doi.org/10.48550/arxiv.2604.26962> (R5717)
- Jin Zhao; Marta Knežević; Tanja Käser (2026). Evaluating Answer Leakage Robustness of LLM Tutors against Adversarial Student Attacks. *arXiv* (Cornell University). <https://doi.org/10.48550/arxiv.2604.18660> (R5765)

- Zheng, C.; Yu, Z.; Jiang, Y.; Zhang, M.; Lu, X.; Jin, J.; Gao, L. (2025). ArtMentor: AI-Assisted Evaluation of Artworks to Explore Multimodal Large Language Models Capabilities. Conference on Human Factors in Computing Systems - Proceedings. Association for Computing Machinery. <https://doi.org/10.1145/3706598.3713274> (R1968)
-

Boundary studies: excluded from analysis

- Frazier, M., Damevski, K., & Pollock, L. (2026). Investigating Conversational Agents to Support Secondary School Students Learning CSP. arXiv:2604.16213. (R5841)
- Liu, A., Sun, M., Esbenshade, L., Tian, V., Zhang, Z., & He, K. (2026). Teacher-Authored Prompts for Configuring Student-AI Dialogue: K-12 Classroom Implementation. arXiv:2604.16738. (R6065)
- Liu, Z.; Yin, S.X.; Lee, C.; Chen, N.F. (2024). Scaffolding Language Learning via Multi-modal Tutoring Systems with Pedagogical Instructions. Proceedings - 2024 IEEE Conference on Artificial Intelligence (CAI 2024), 1258-1265. Institute of Electrical and Electronics Engineers. <https://doi.org/10.1109/cai59869.2024.00223> (R1662)